

Reaction types in Organometallic Chemistry

—Elementary Reactions in Organometallic Chemistry (基元反应)

- (1) Ligand coordination and substitution ✓
(配体的配位和取代)
- (2) Oxidative addition and reductive elimination ✓
(氧化加成和还原消除)
- (3) Insertion and elimination (de-insertion) reactions ✓
(插入反应和消除(反插入)反应)
- (4) Electrophilic and Nucleophilic Addition and Abstraction
(亲电和亲核加成与攫取)
(Reactions on the complexed ligands 和金属结合的配体的反应)

Help us to understand and predict the metal-catalyzed or mediated reactions.

Be aware:

- Classified **in terms of reaction patterns**, not mechanism !!
- An elementary reaction may go through different mechanisms.

Chapter 8. Electrophilic and Nucleophilic Addition and Abstraction (Reactions of the complexed ligands)

Outline

- A. Introduction**
- B. Nucleophilic addition to CO and CNR**
- C. Nucleophilic addition to olefins and acetylenes**
- D. Nucleophilic abstraction**
- E. Electrophilic addition**
- F. Electrophilic abstraction**

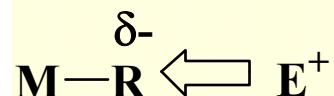
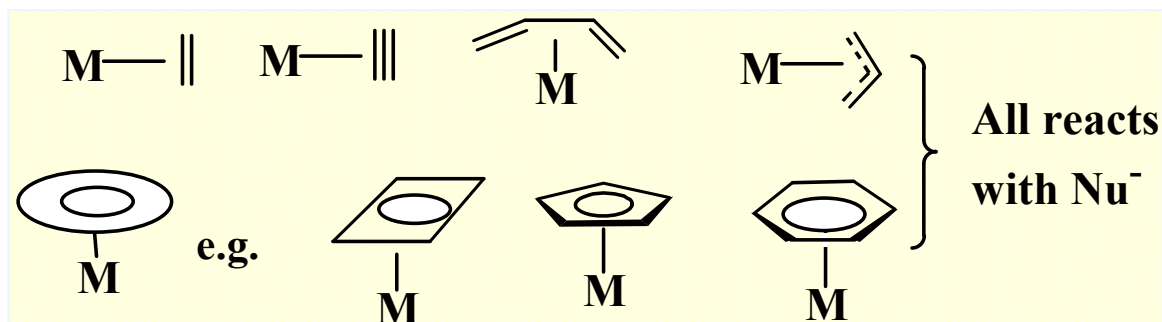
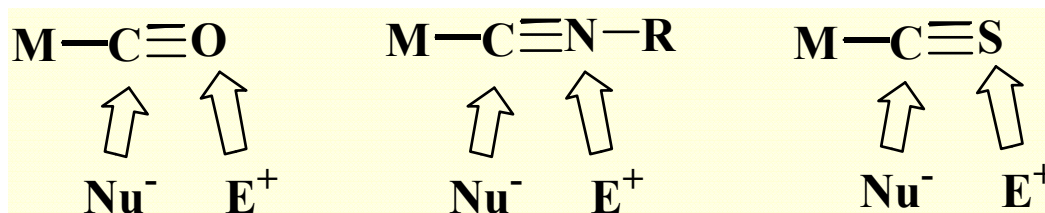
References and suggested readings:

1. *The Organometallic Chemistry of the Transition Metals*, Robert H. Crabtree, 6th Edition, 2014, Chapter 8.
2. *Organotransition Metal Chemistry*, Akio Yamamoto, 1986. 272-286

A. Introduction

Organic ligands are activated upon complexation. They may react with electrophiles or nucleophiles.

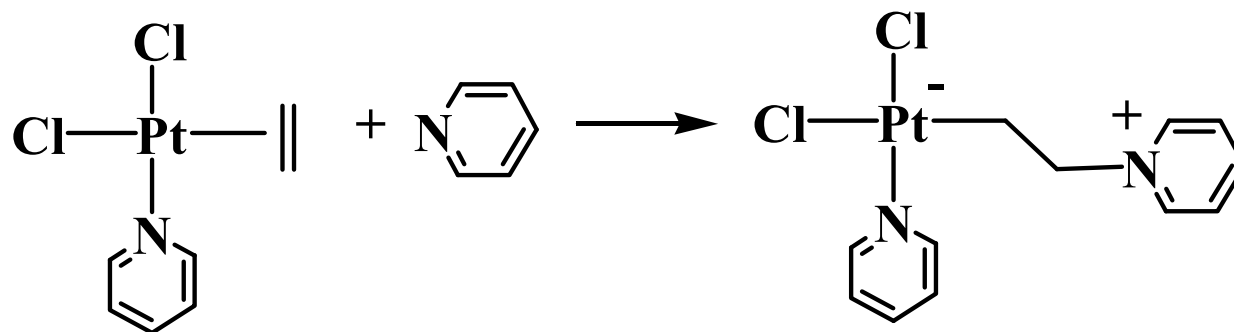
e.g. We have shown before:



Two possible outcomes for the reactions with nucleophiles

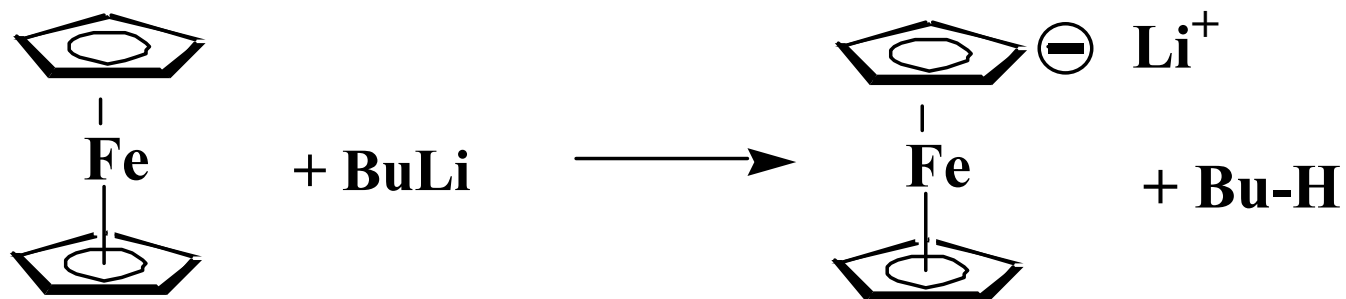
(1) **Nucleophilic addition** - reagent stays on metallic fragment

e.g.



(2) **Nucleophilic abstraction (攫取)** - reagents do not stay with metallic fragment

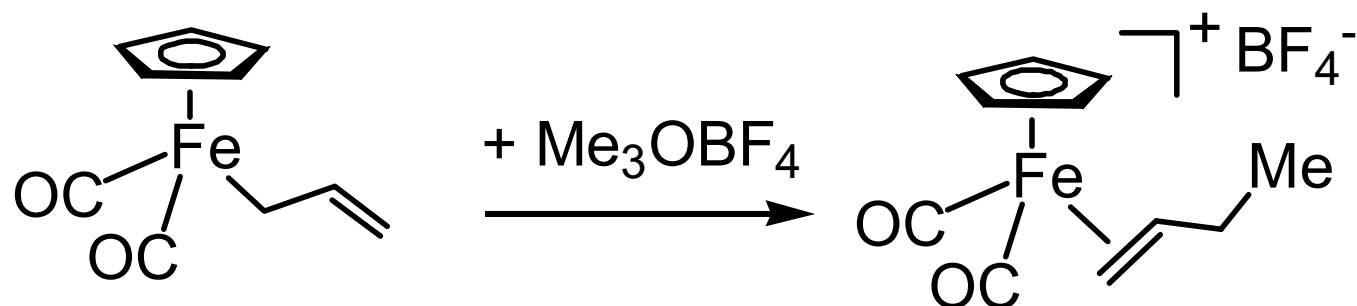
e.g.



There are also two possible outcomes for reactions with electrophiles.

(1) **Electrophilic addition** - reagent stays with metallic fragment.

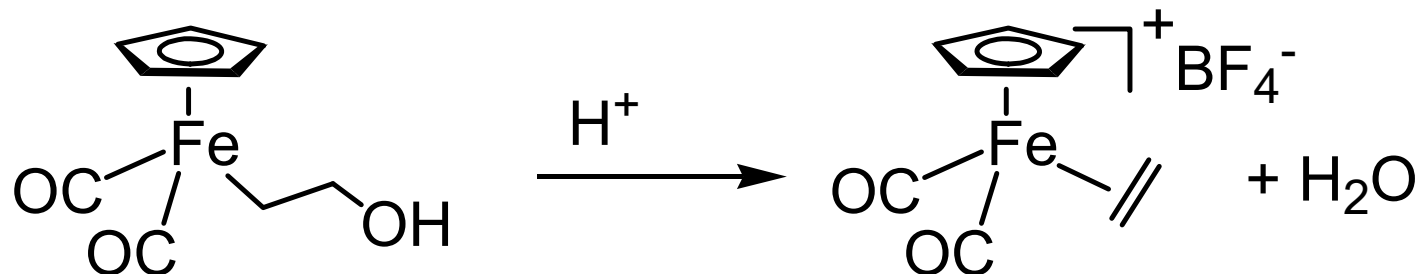
e.g.



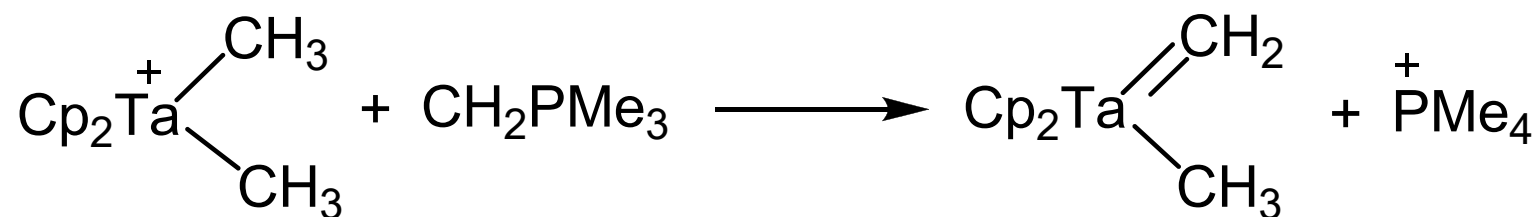
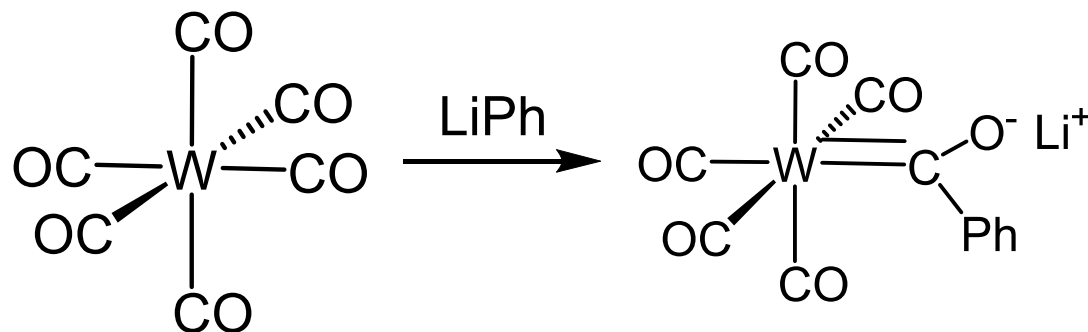
(Me_3OBF_4 is used to supply Me^+ .)

(2) **Electrophilic abstraction** - reagent does not stay with metallic fragment.

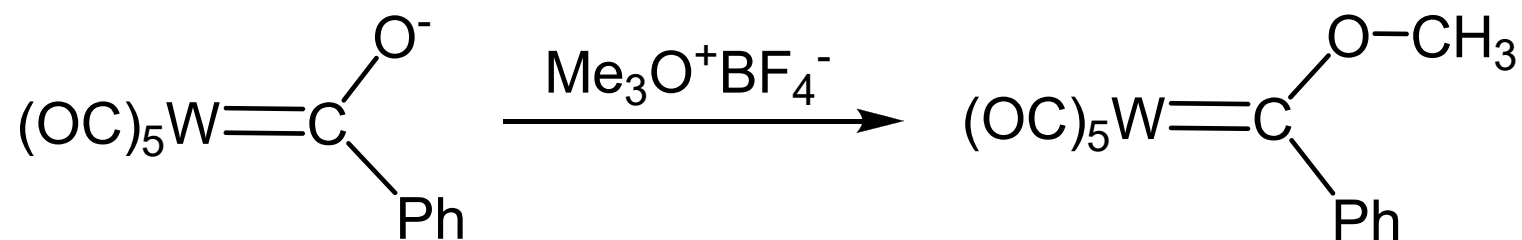
e.g.



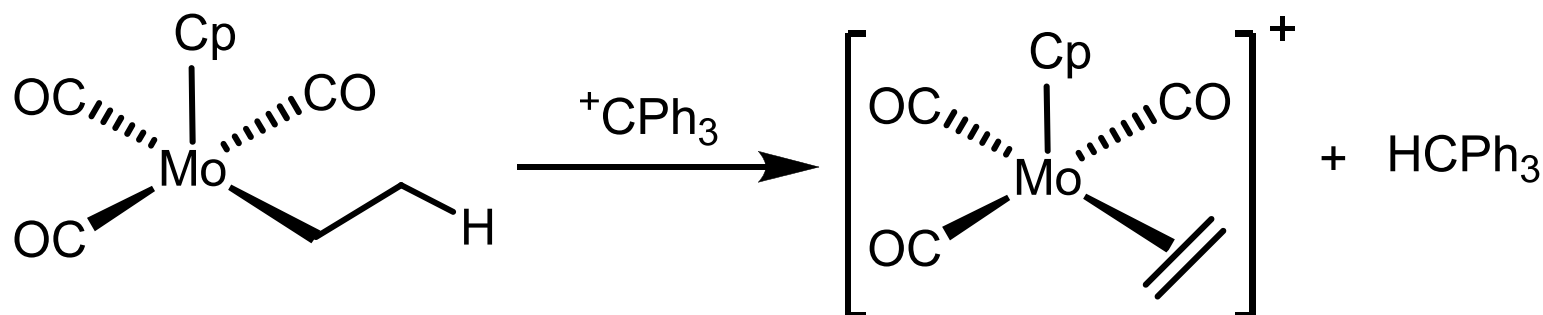
Nucleophilic attack is favored for complexes with net positive charges or with π -accepting (π -acid) ligands.



Electrophilic attack is favored for electron rich metal complexes, *i.e.* low valent metals, or those with net negative charges and/or those with strong electron donating ligands.

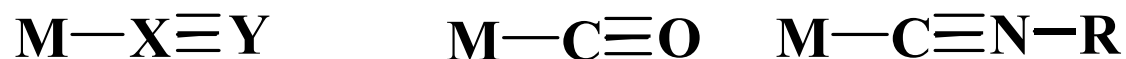


Electrophilic addition



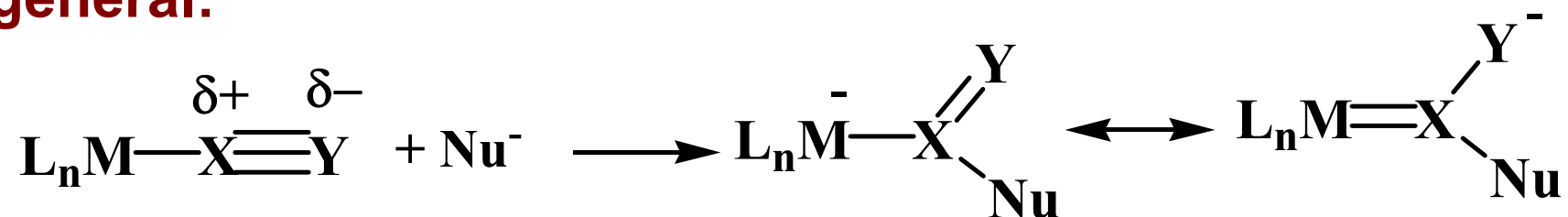
Electrophilic abstraction

B. Nucleophilic addition to CO and related ligands



We have mentioned before that *M-CO*, *M-CNR* and *M-CS* can be attacked by *Nu⁻* (at the α -position), especially positively charged species.

In general:

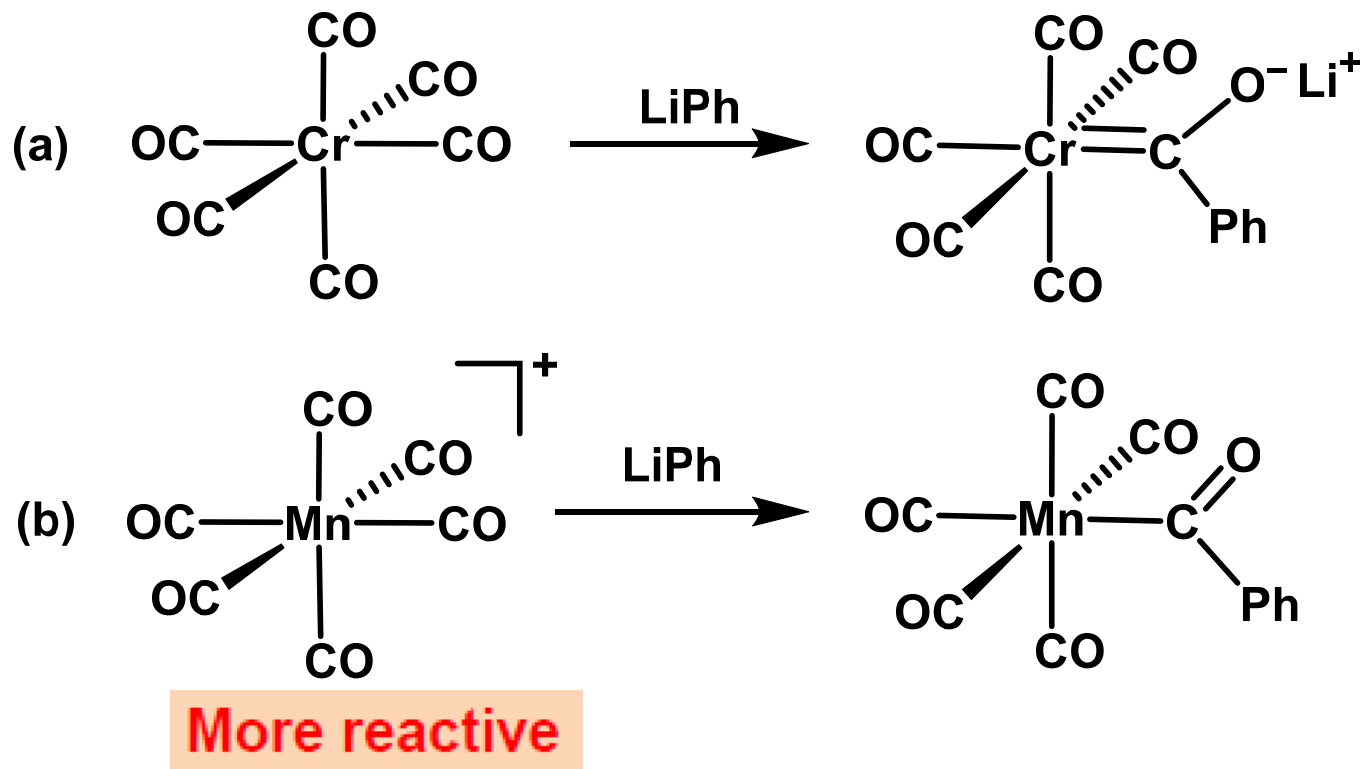


Reactivity: CS > CNR > CO > NO > N₂

The reactivity is affected by the electron-richness of the metal center.

Review questions

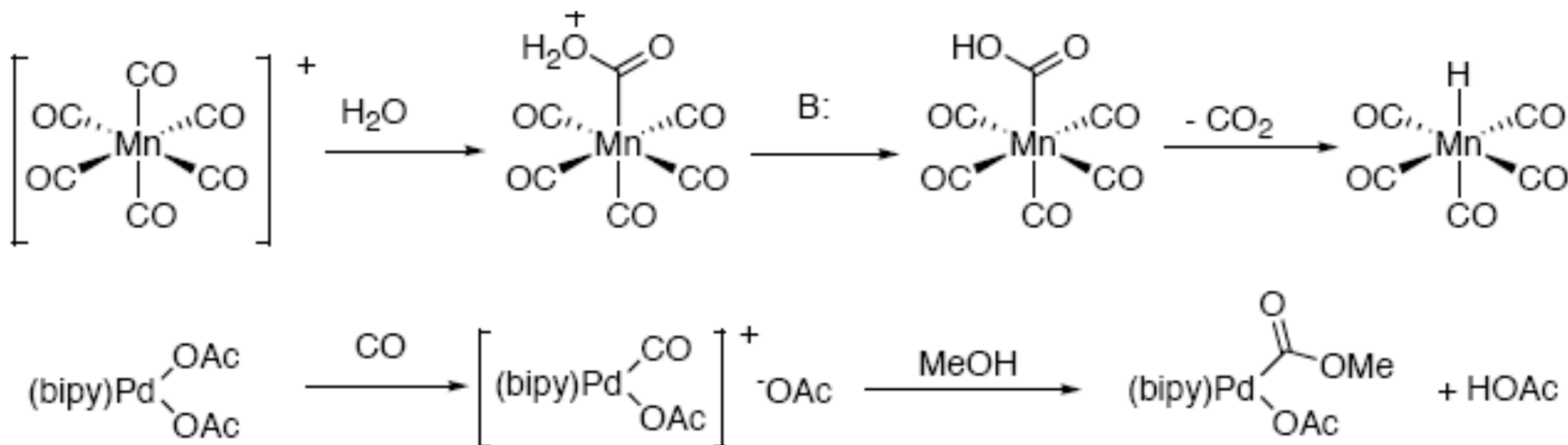
Q1. Consider the reactions below, which one would you expect to be faster?



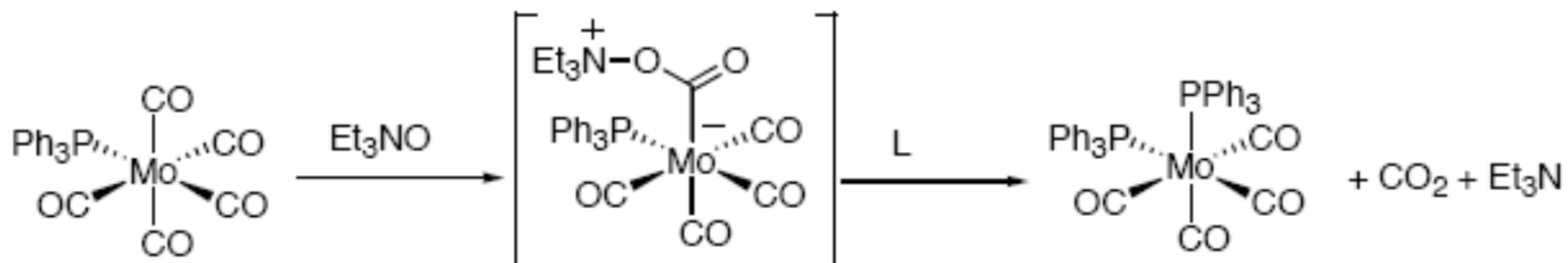
Answer: (b) (Positive charge species)

Electron deficient complex reacts faster.

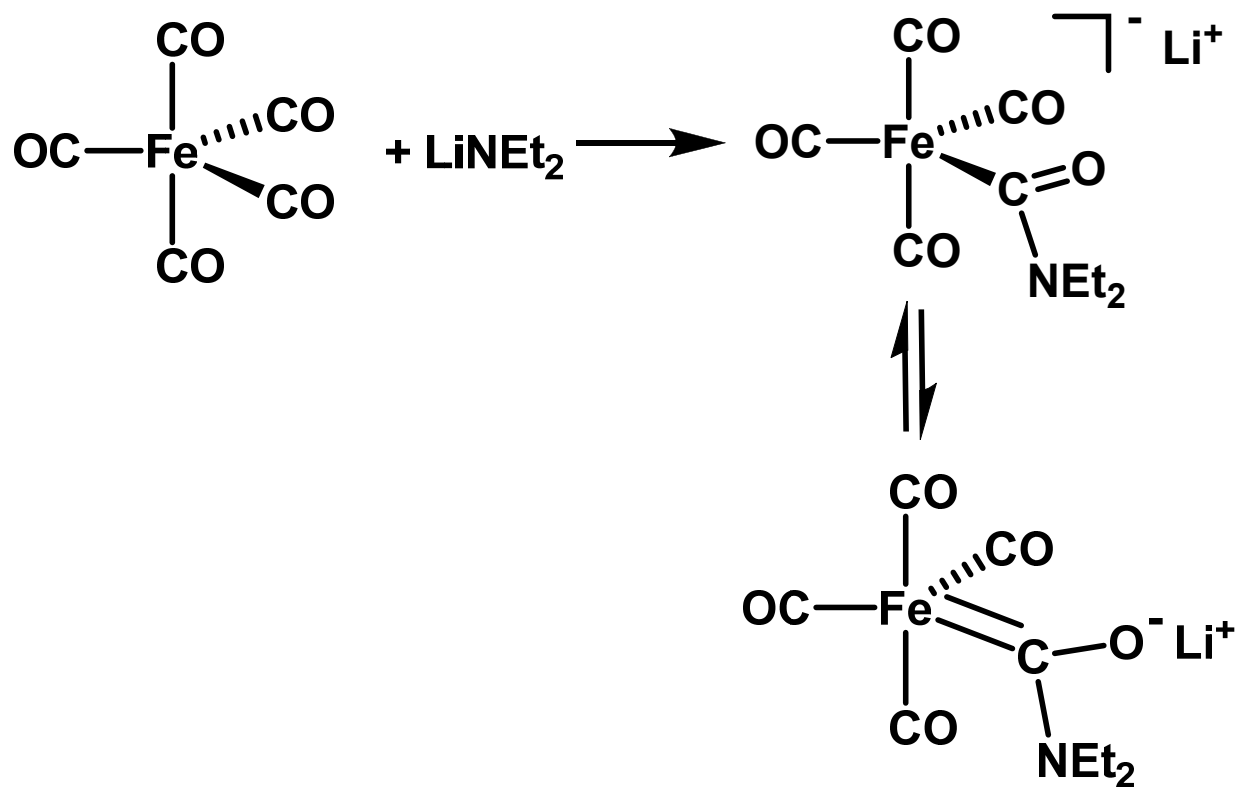
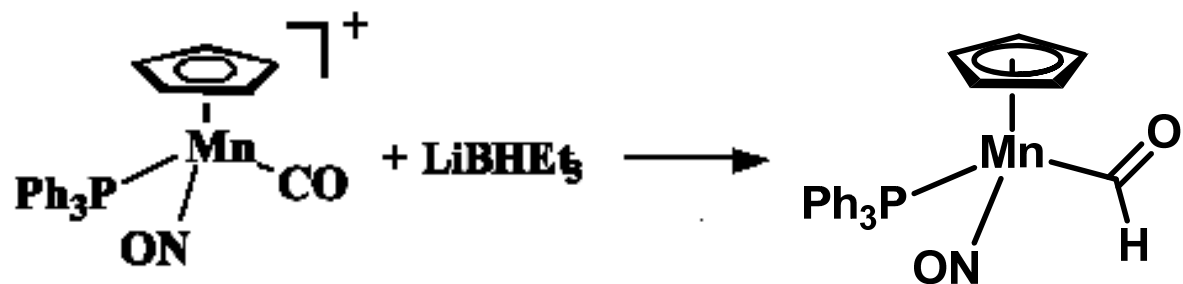
Increasing the electrophilicity of the metal center allows weaker nucleophiles like water or alcohols to attack.

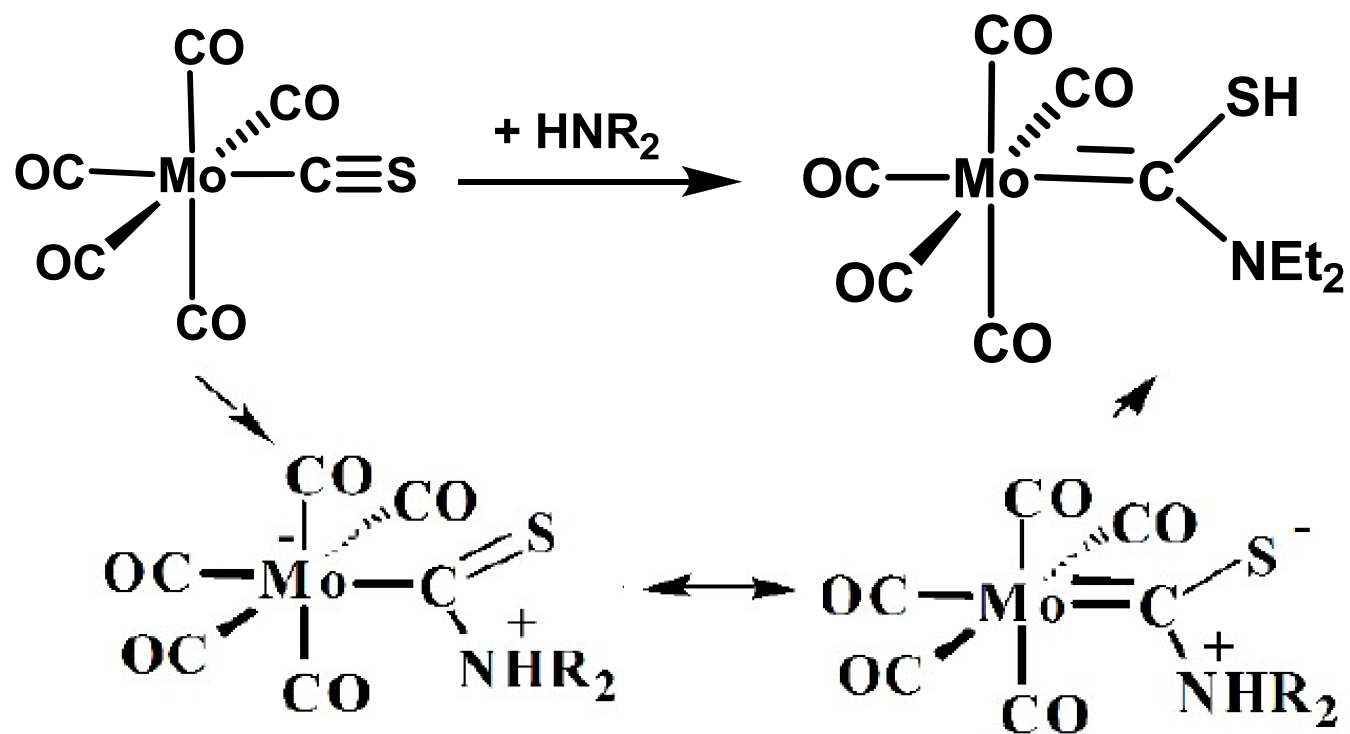
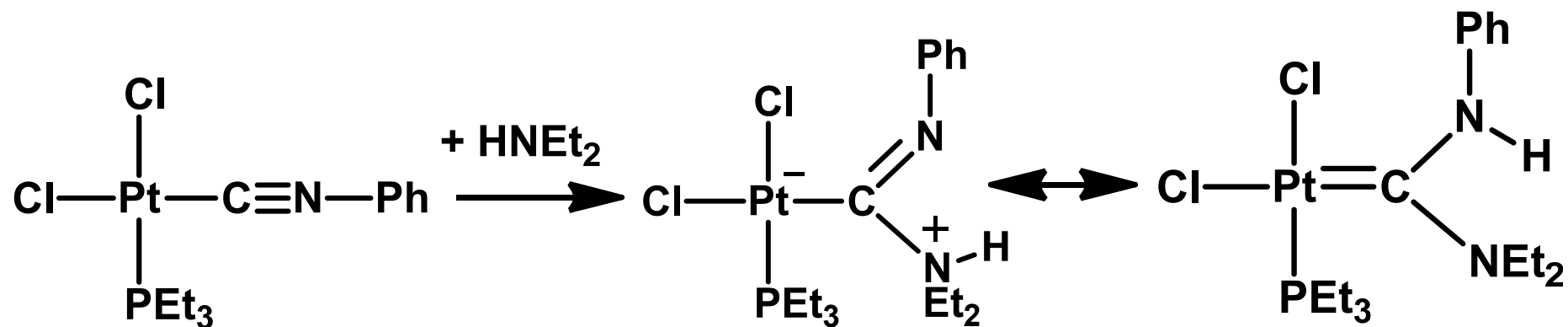


As discussed before, *N-oxides are strongly nucleophilic reagents that can be used to abstract CO ligands. The ligand substitution occurs exclusively cis to the phosphine.*

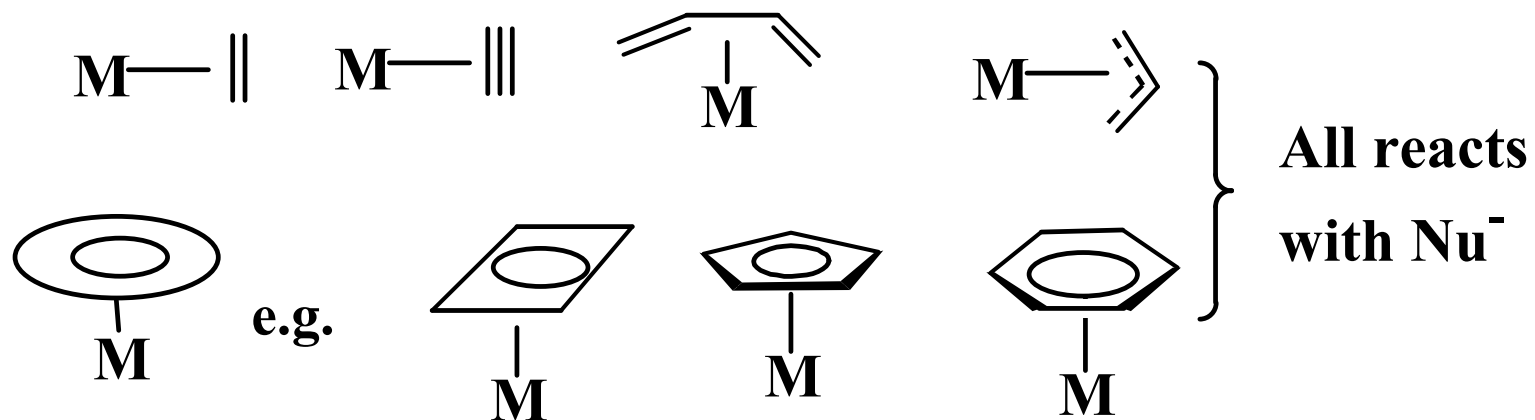


Q2. Provide the product of the following reactions



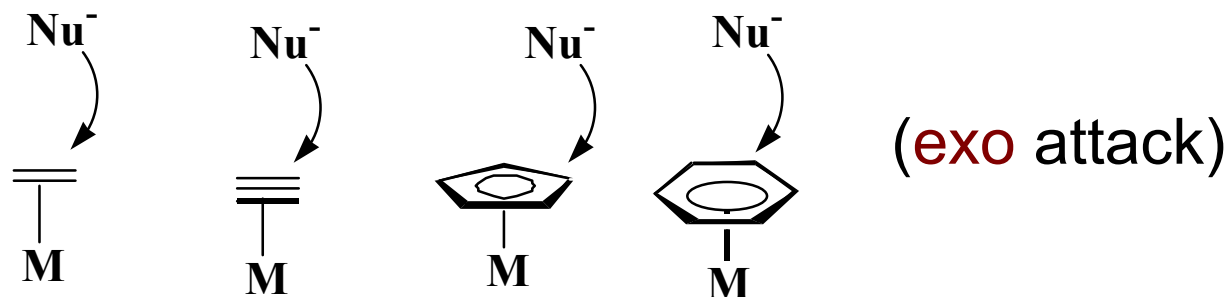


C. Nucleophilic addition to coordinated olefins and acetylenes



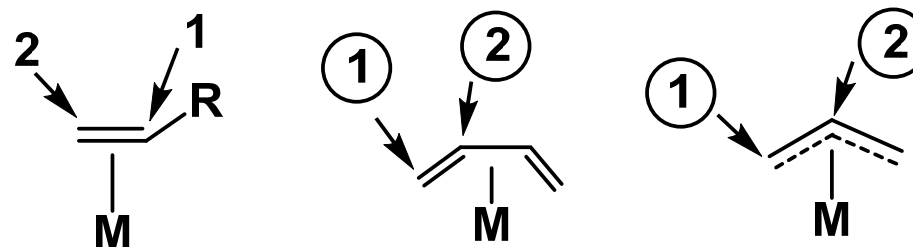
Most common features

- Direction of attack?** Attack from the opposite side of M



- Regio chemistry?**

e.g. attack site 1 or 2?

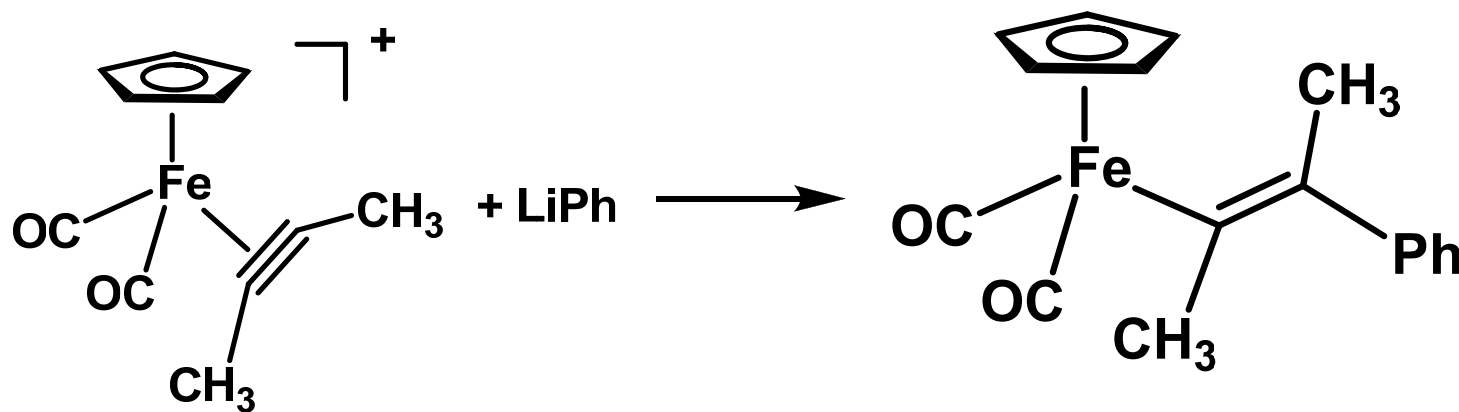


Attacked site

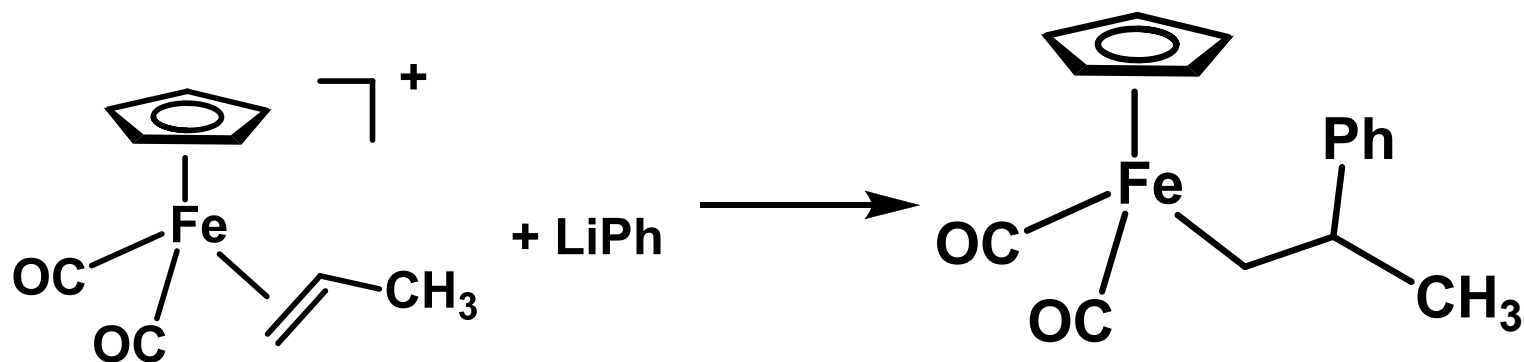
Site 1 is Preferred			
Less common			

Practice Problems

Give the products of the following reactions.

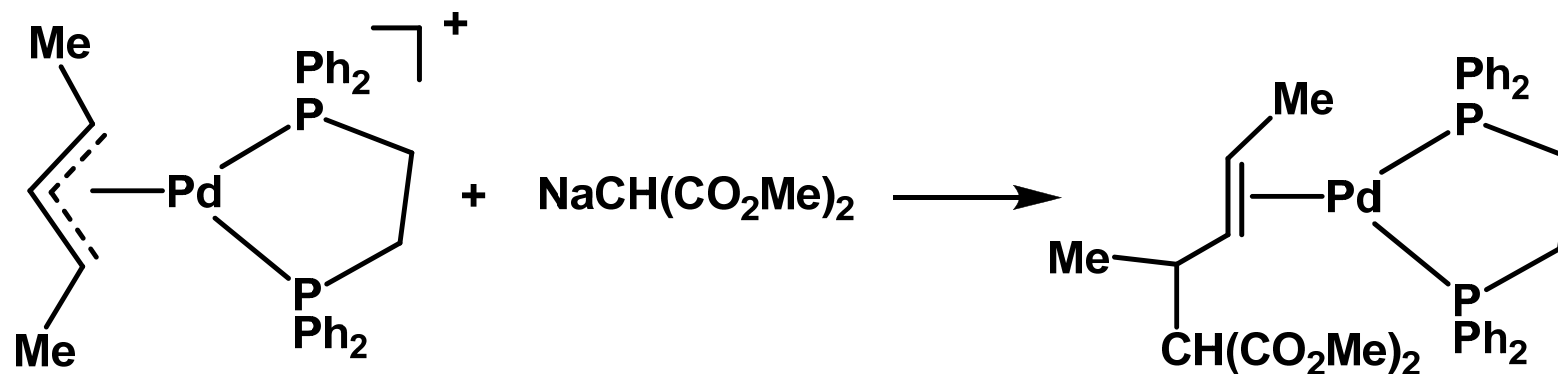
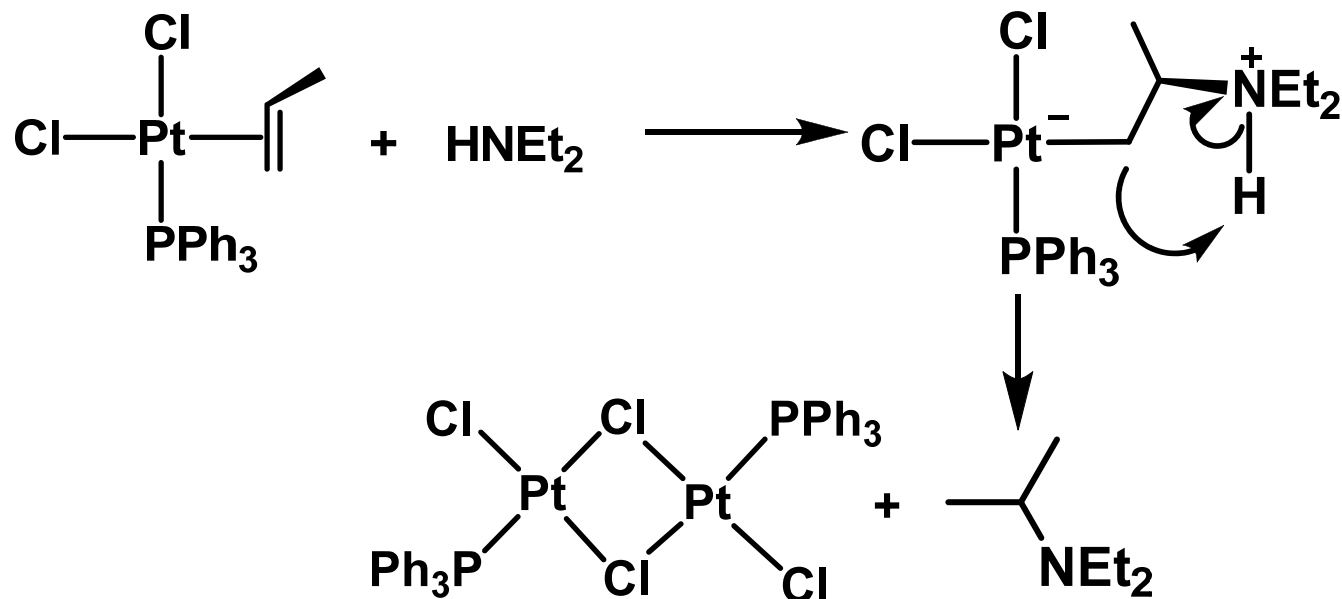


The stereochemistry is *anti* for the two substituents of $\text{C}=\text{C}$.
(different from the case in *Insertion Reaction*)



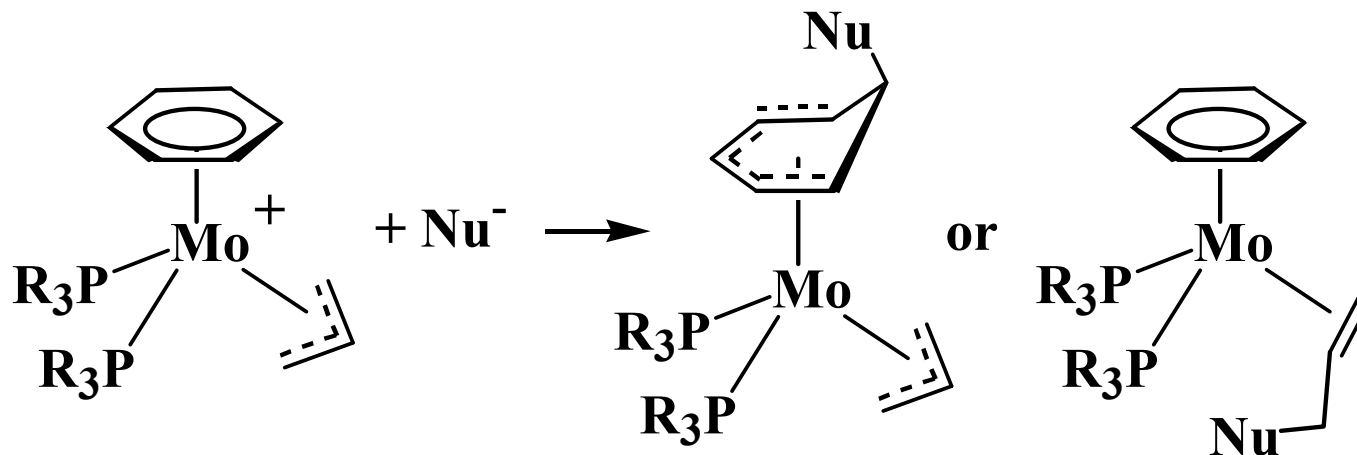
Practice Problems

Give the products of the following reactions.



- **Site of selectivity.**

If there are more than one type of olefins bonded to a metal, which one will be attacked? *e.g.*



How to determine the site selectivity?

There are several rules: Davies-Green-Mingos Rules

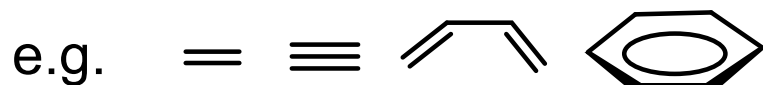
In a complex with several polyene or polyenyls, we often see selective attack at one site only. Davies, Green, and Mingos systematized these reaction outcomes in terms of rules that usually correctly predict the site of addition.

(经验规则，通常能正确地预测加成反应发生的位置)

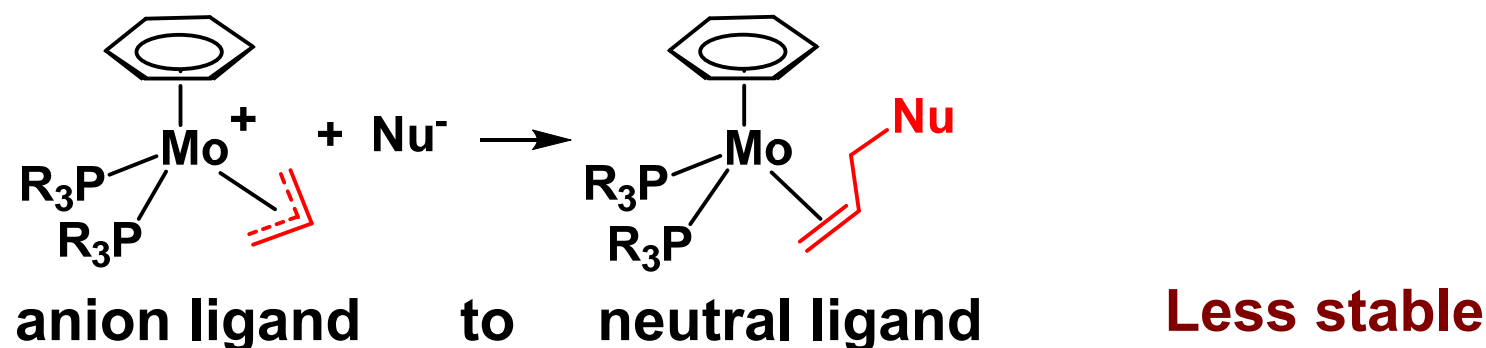
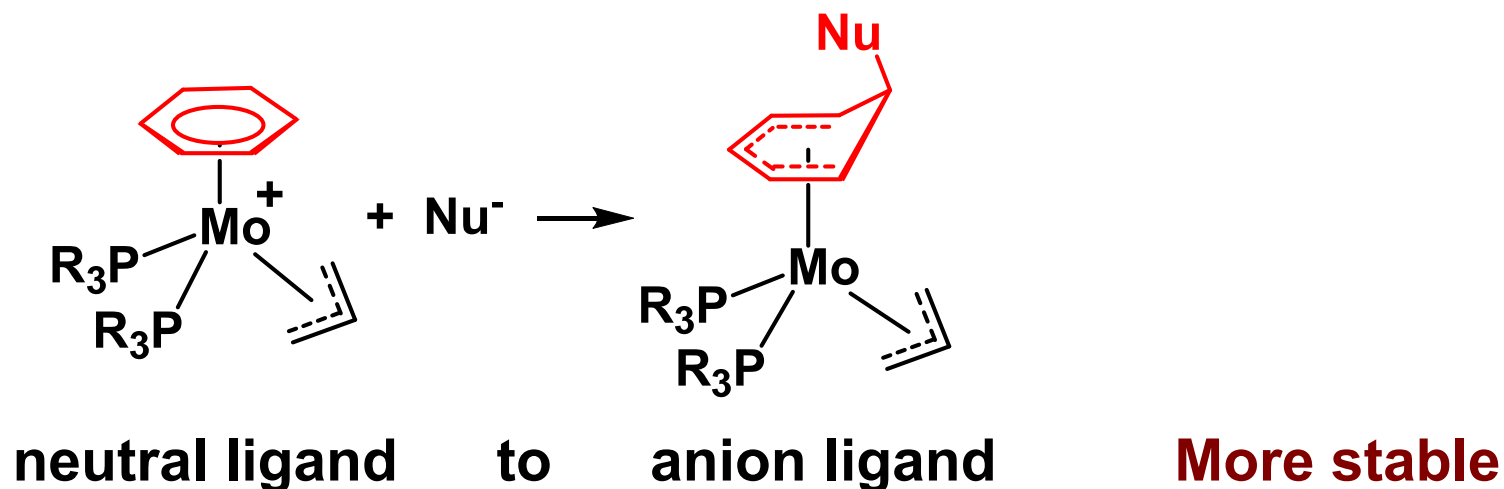
Davies-Green-Mingos Rules

Rule 1. Polyenes reacts before polyenyls

Polyenes (even # C)

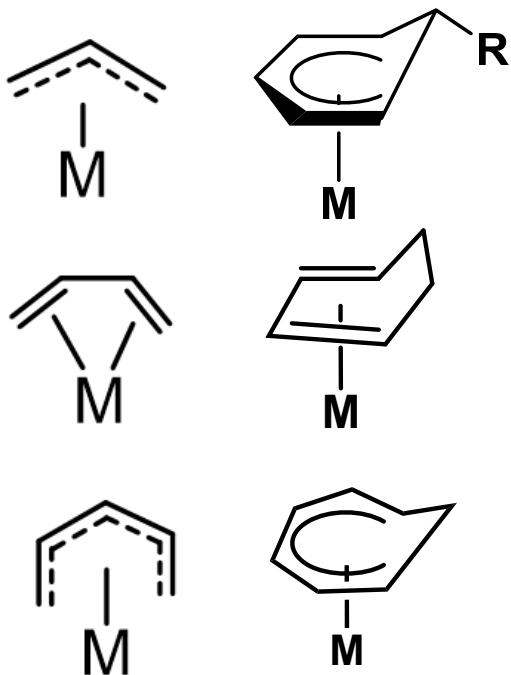


Polyenyls (odd # C)

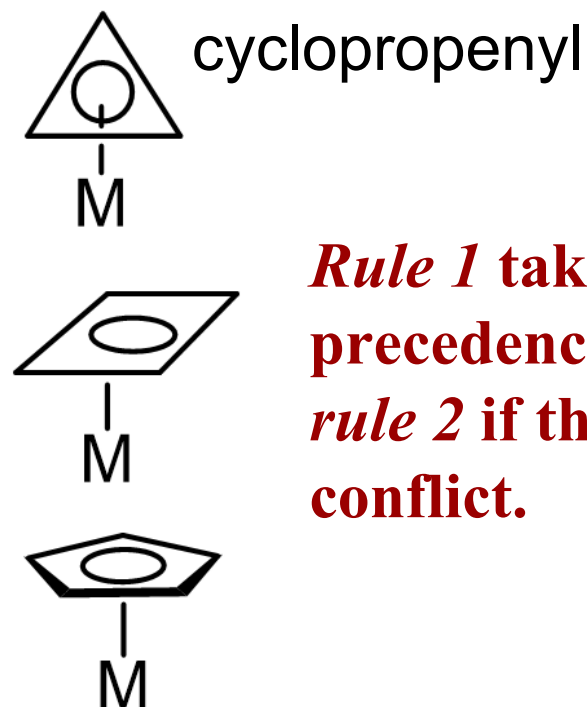


Rule 2. Open ligand reacts before closed ligands

Open ligand
(with interrupted
conjugation)

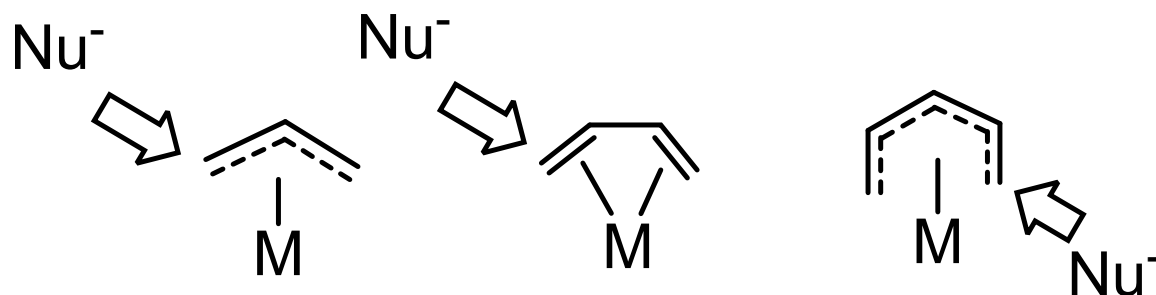


Closed ligand
(with cyclic conjugation)



***Rule 1 takes
precedence over
rule 2 if they
conflict.***

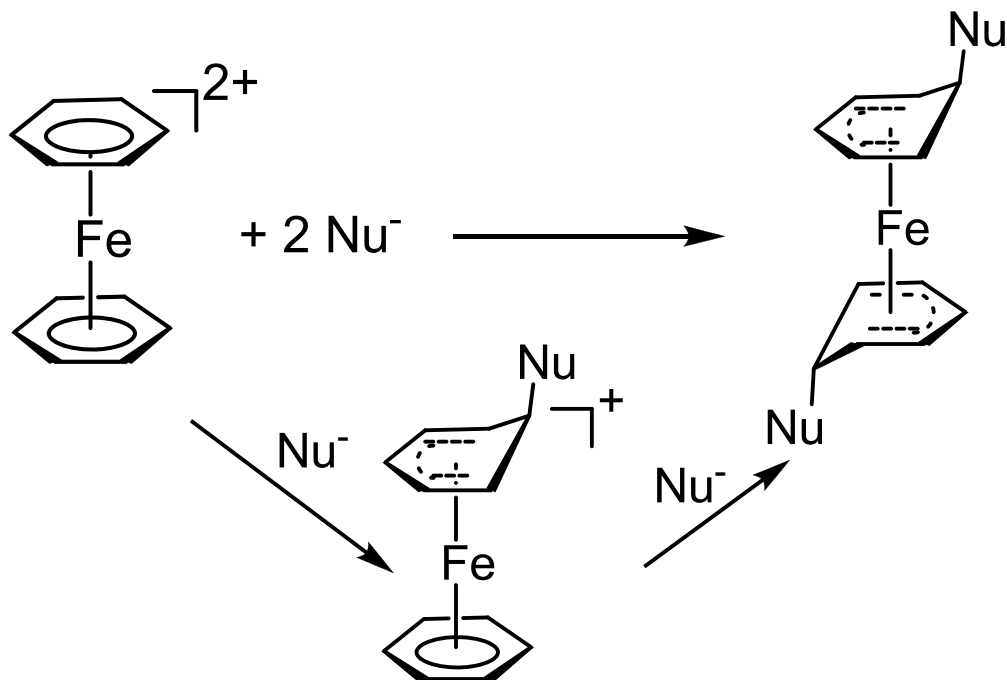
Rule 3. Usually terminal addition



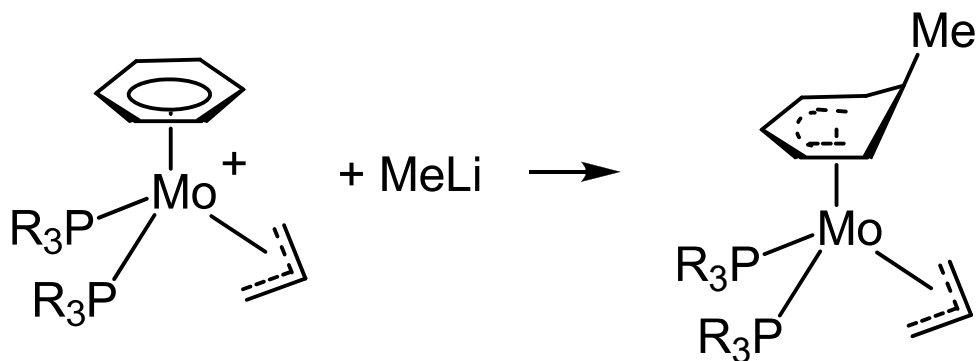
- Open polyenes give terminal addition.
- Open polyenyls usually give terminal attack
(but non-terminal if L_nM is particularly electron donating.)

Practice Problems

Give the products of the following reactions.

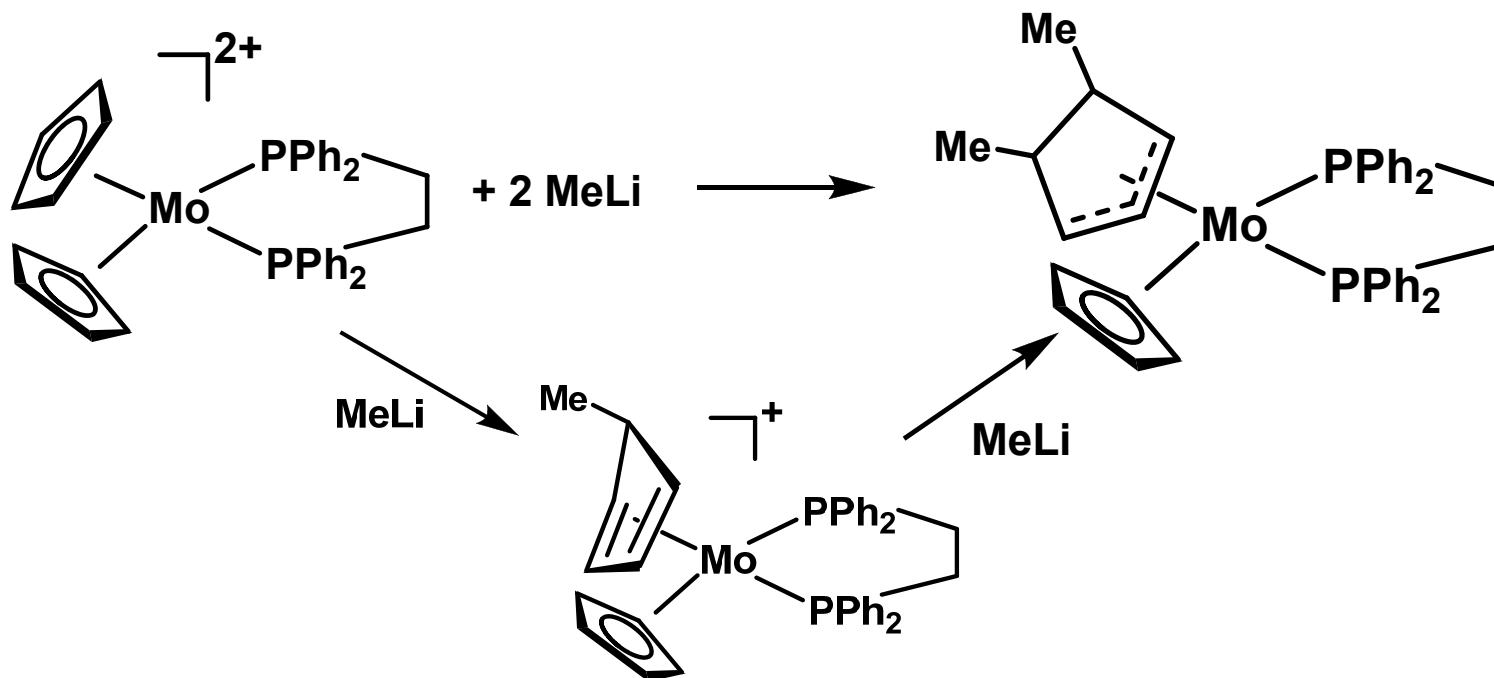


A second nucleophile can also add, but to the other ring, as predicted by Rule 1.



First attack the even closed arene rather than the odd open allyl

Give the products of the following reaction.

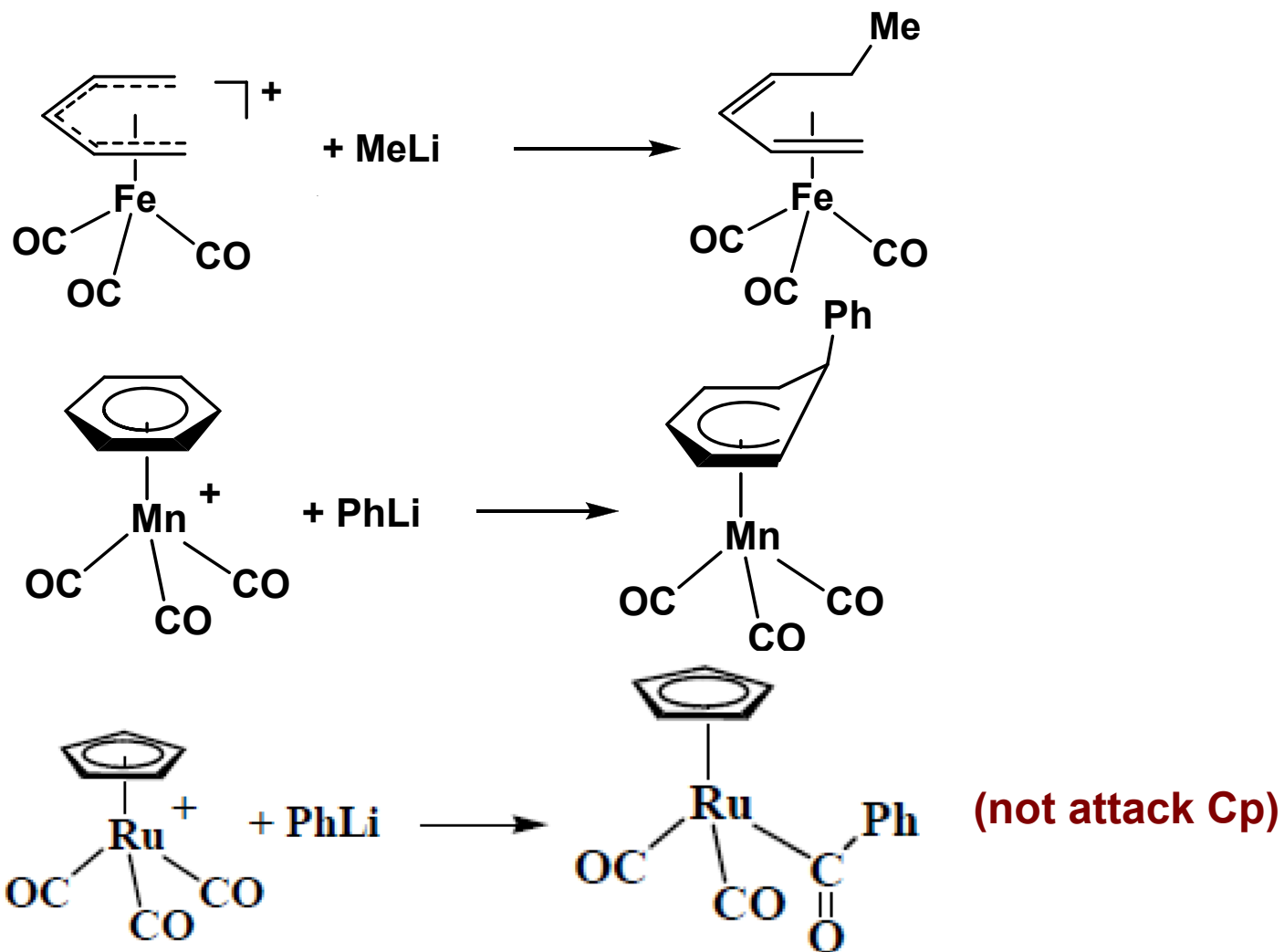


As an **odd closed system**, Cp is usually very resistant to attack and therefore directs addition to other ligands present on the M.

Attack at a Cp ring only happens if there is no other π -bonded ligand present.

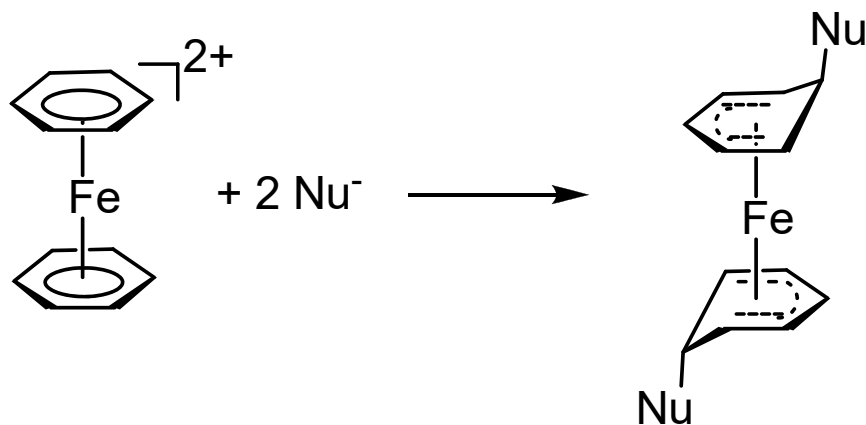
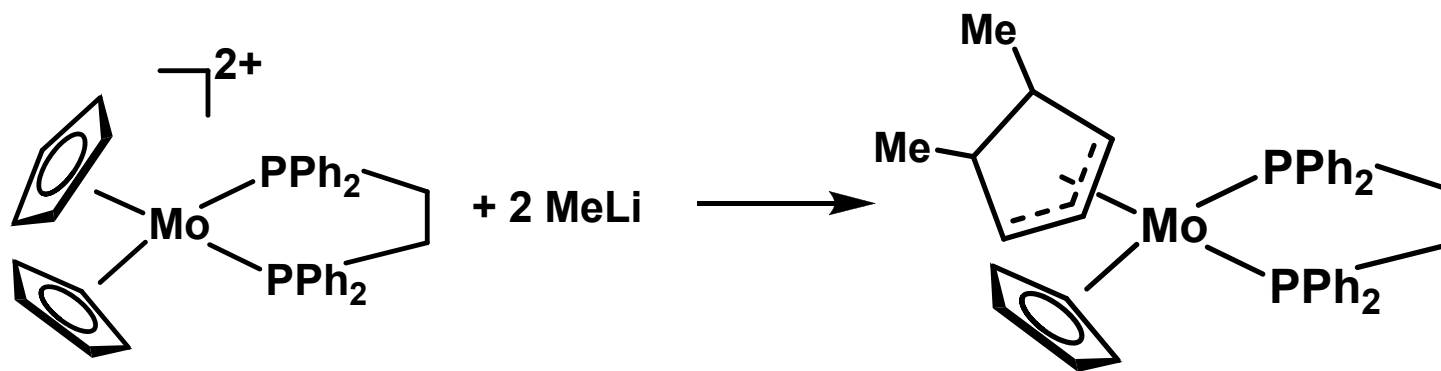
In general, Cp is a stabilizing ligand in studies on nucleophilic attack.

CO is an even ligand but, not being π -bound, is among the least reactive of these.



Reactivity: ene > open enyl > CO > closed enyl

Rule 4. “Zeroth rule” of nucleophilic addition



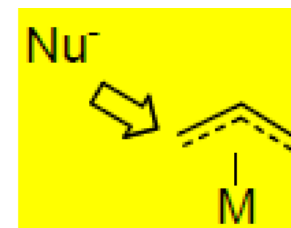
A second nucleophile can also add, but should obey Rule 1.

“zeroth rule” of nucleophilic addition: A nucleophile usually adds once to a monocation, twice to a dication, and so on.

Some exceptions

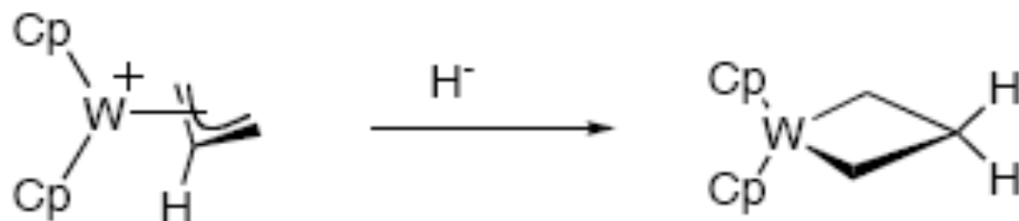
For allyl:

✓ the attack on central carbon may also occur.

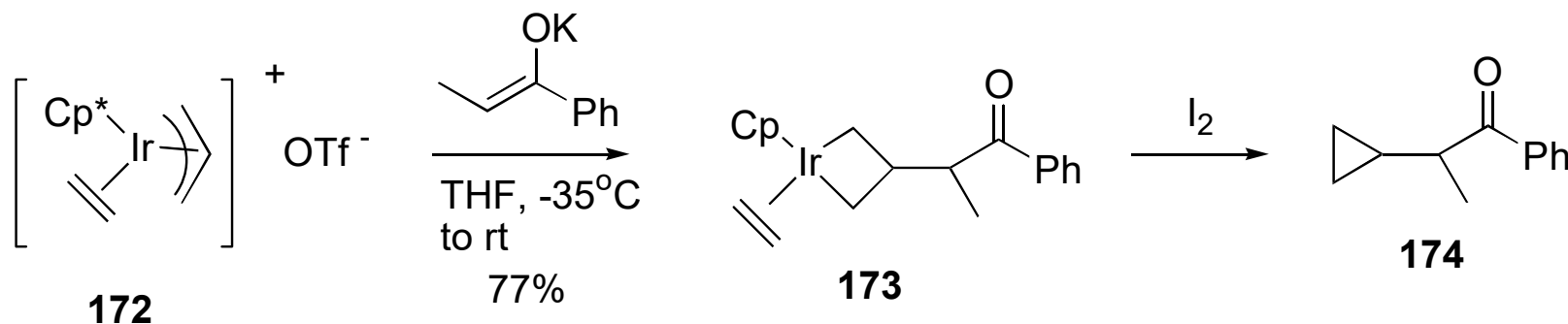
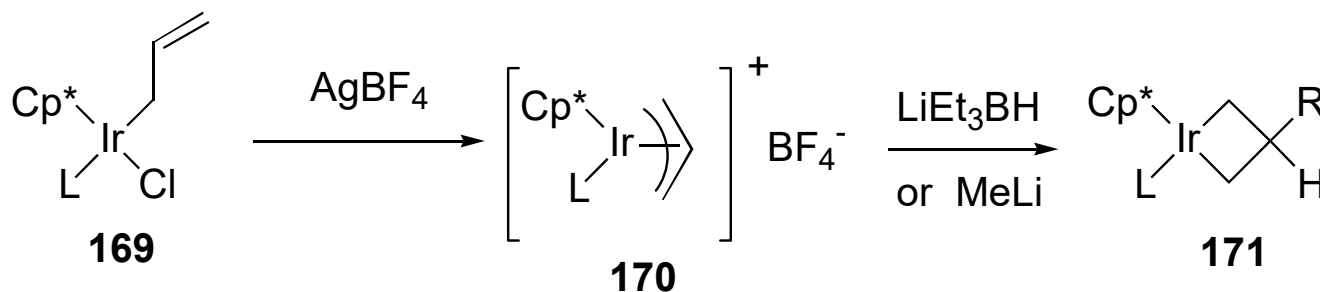


Usual reactivity

Particularly when the L_nM is strong electron donating fragment.

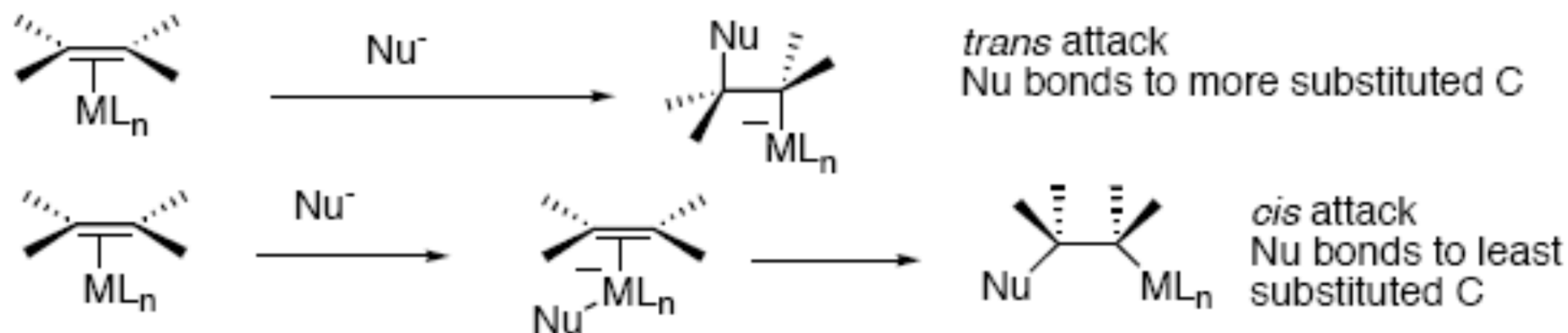
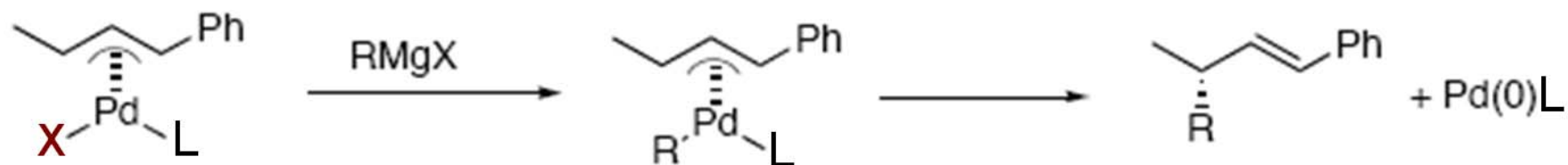


Cp_2W^+ (d^2 fragment) is strongly electron donating



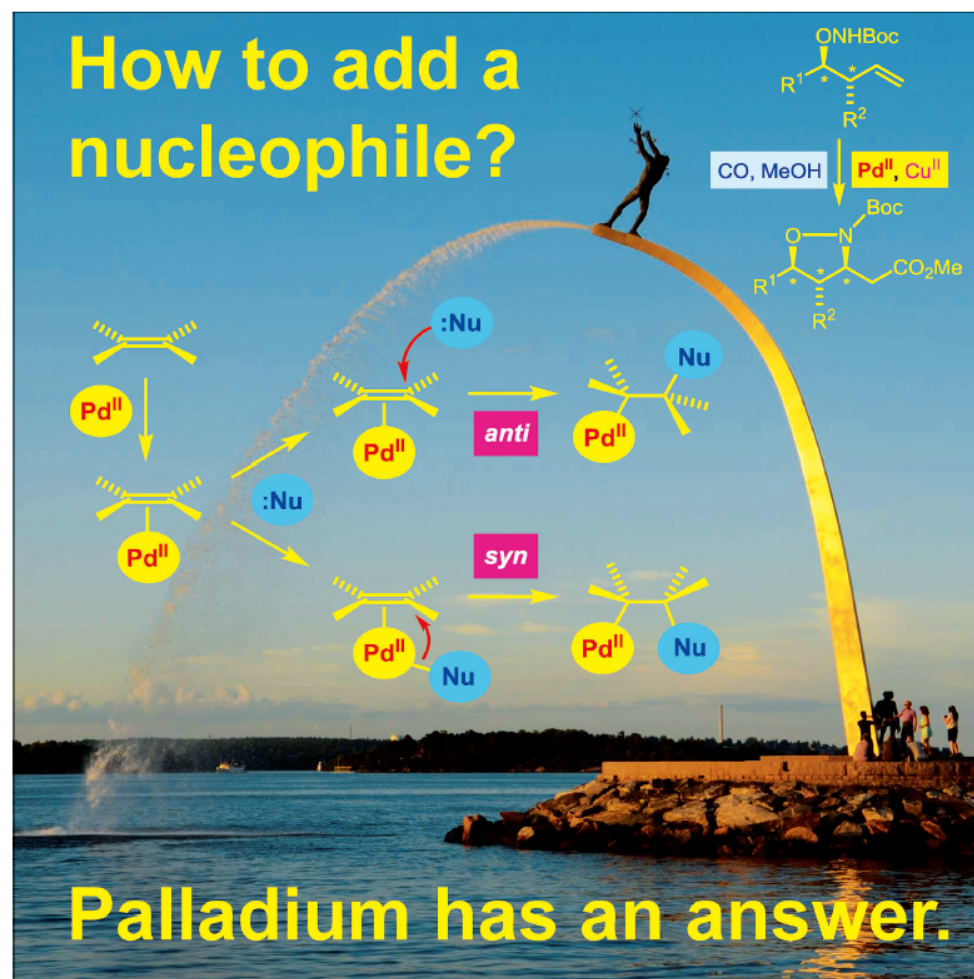
Some exceptions

Soft nucleophiles may also **attack on the metal initially**, followed by migratory insertion. e.g.



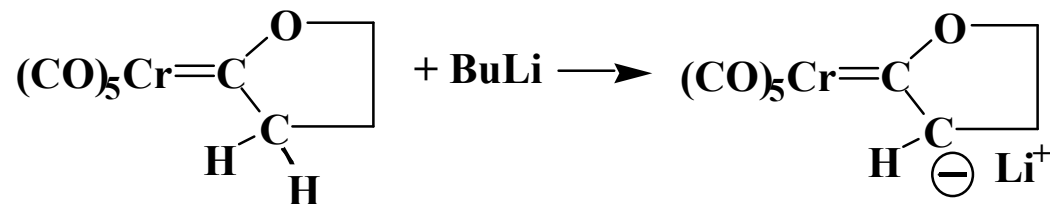
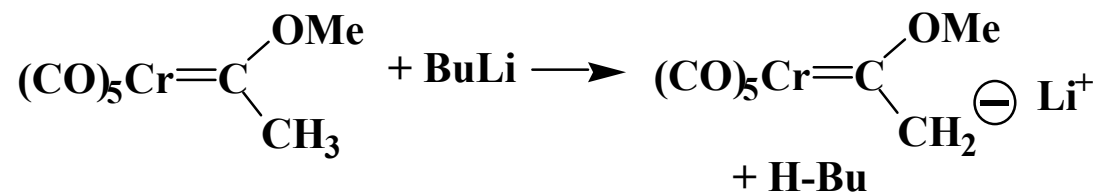
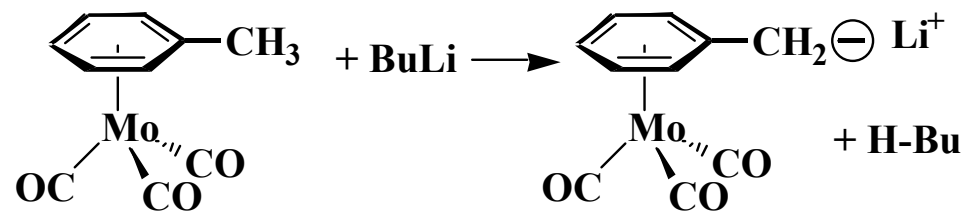
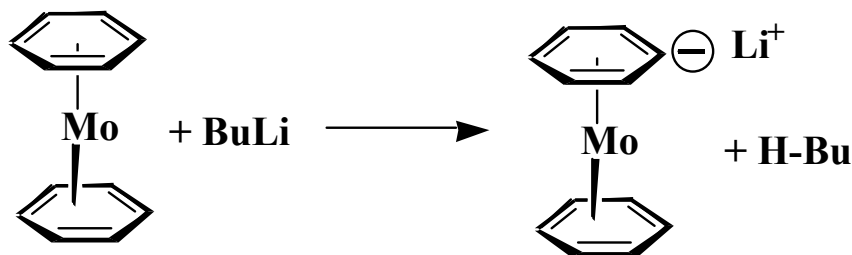
The syn/anti-Dichotomy in the Palladium-Catalyzed Addition of Nucleophiles to Alkenes

Pavel Koc̣ovský and Jan-E. Bäckvall*



D. Nucleophilic abstractions.

Some Organometallic complexes have acidic protons and can be deprotonated. e.g.

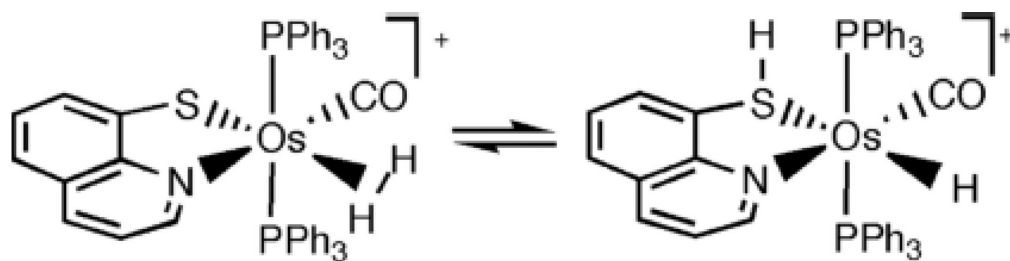
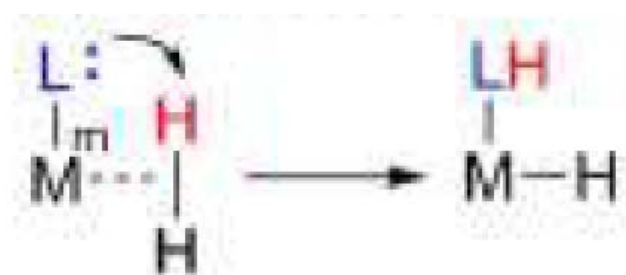


D. Nucleophilic abstractions.

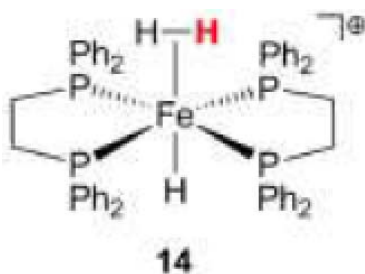
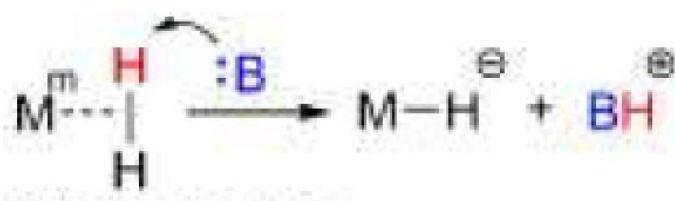
Additional examples:

- ✓ Deprotonation of dihydrogen complex

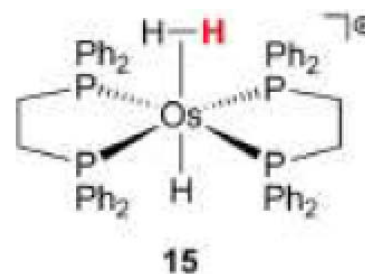
Intramolecular



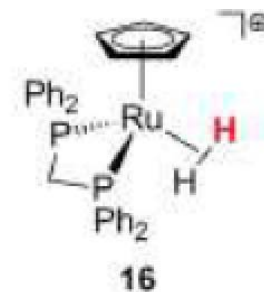
Intermolecular



pKa: 14.3



14.1



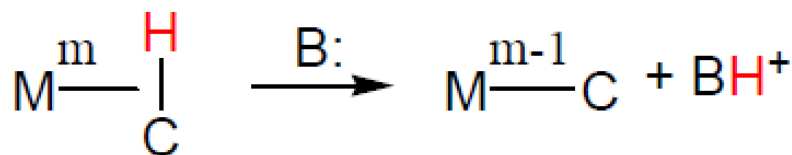
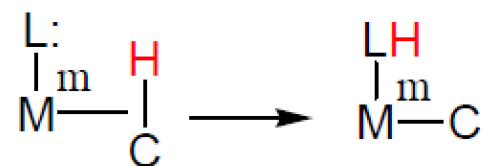
9.1

D. Nucleophilic abstractions.

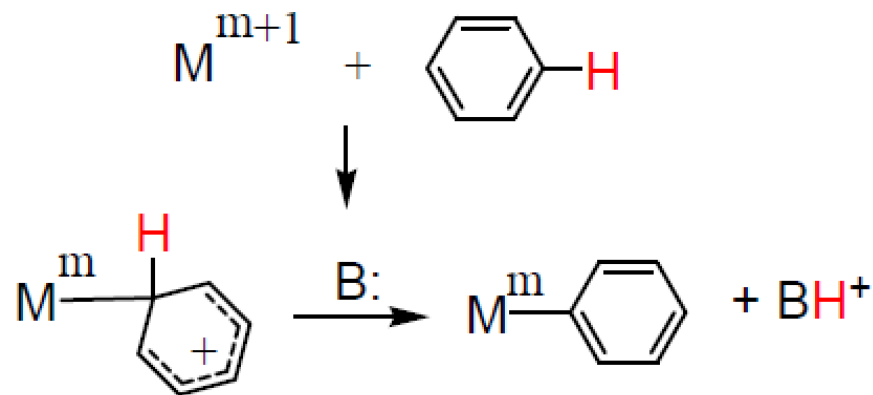
Additional examples:

- ✓ Base assisted C-H activation: metalation-deprotonation

Intramolecular



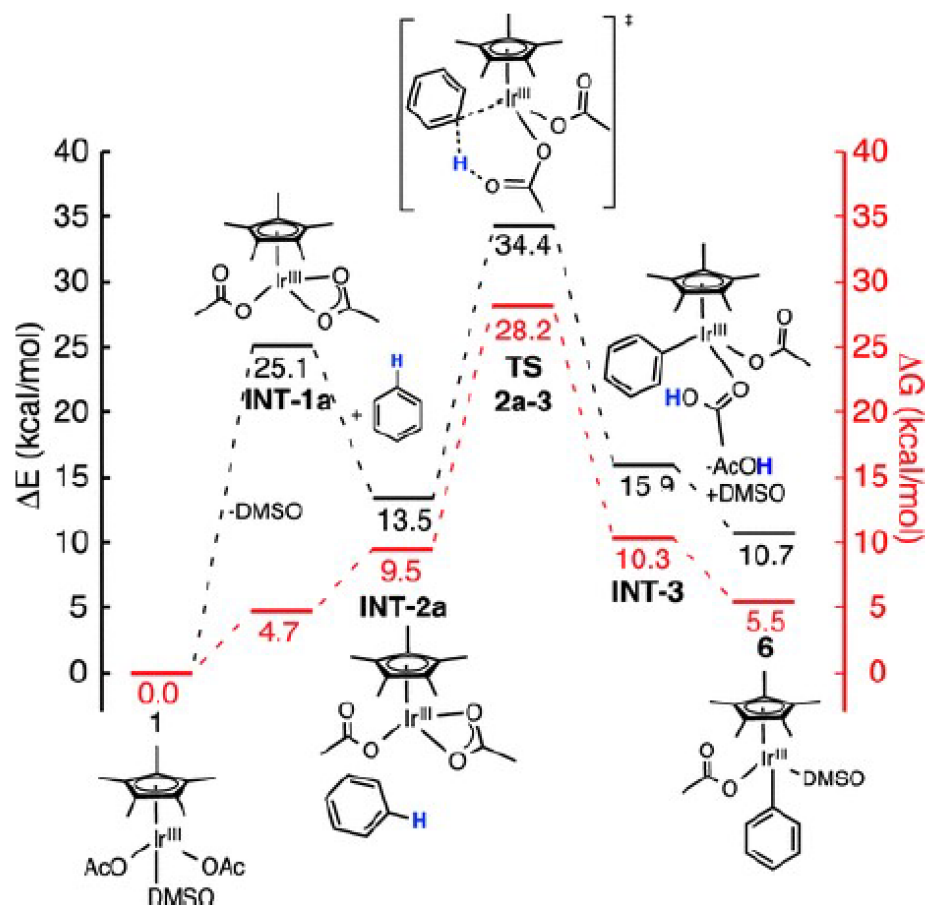
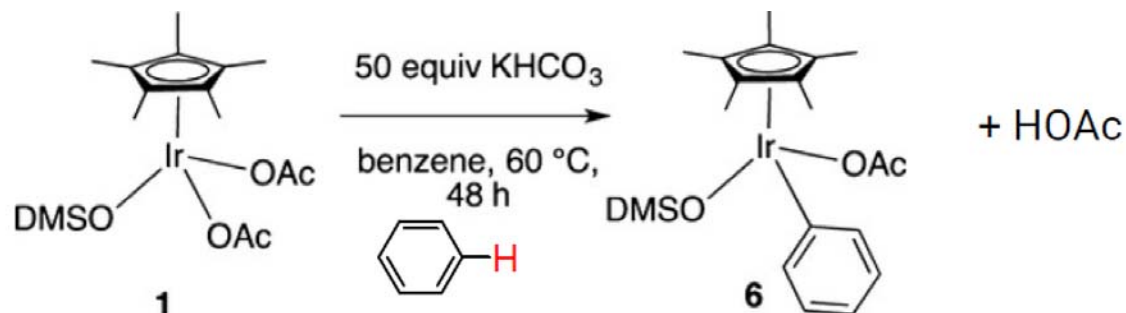
Intermolecular



D. Nucleophilic abstractions.

✓ Base assisted C-H activation

e.g.



Concerted Metalation-Deprotonation (CMD)

Relative electronic energies are in black, and the relative free energies are in red.

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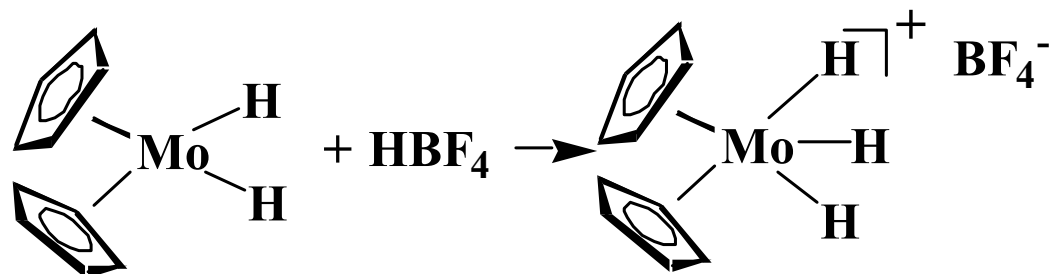
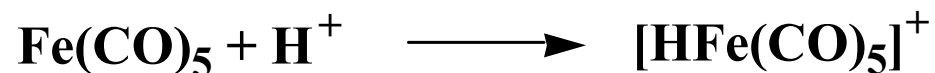
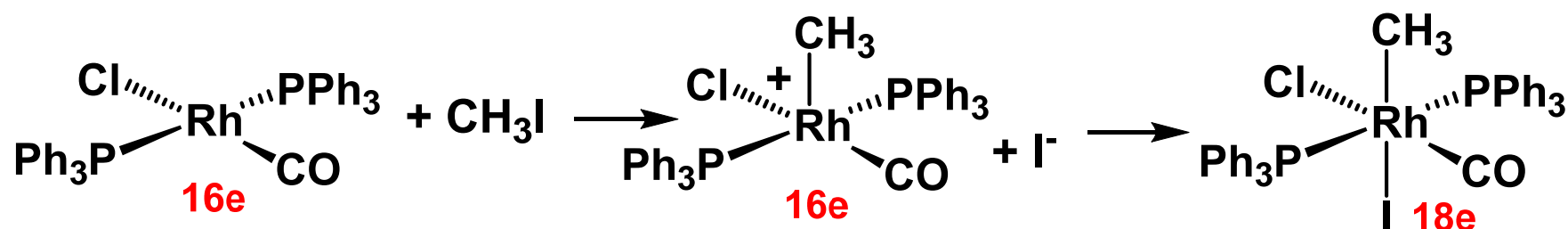
E. Electrophilic addition

(1) Electrophilic addition to metals.

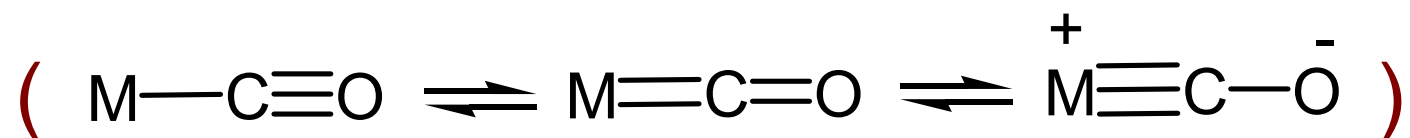
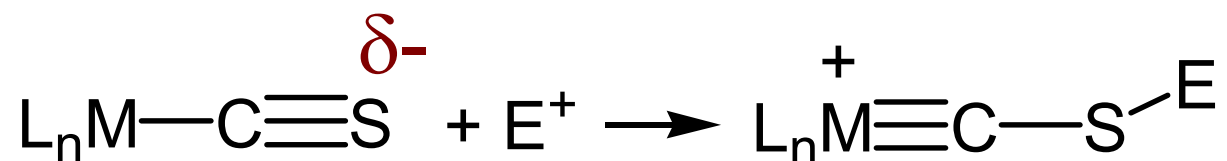
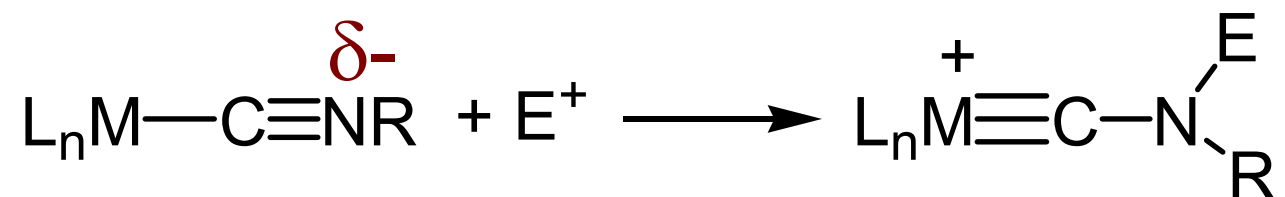
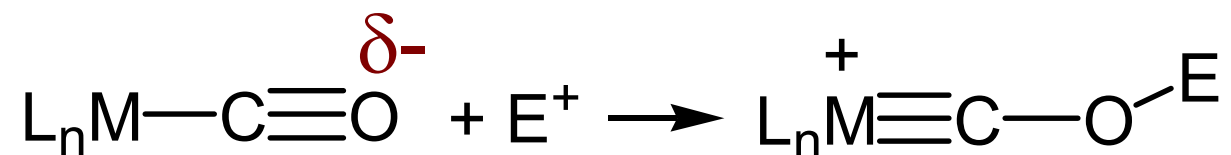
(Special oxidative addition reaction)



e.g.

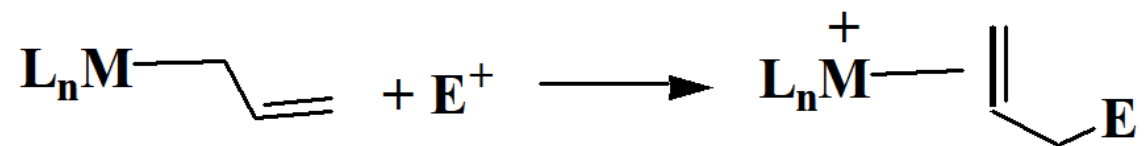
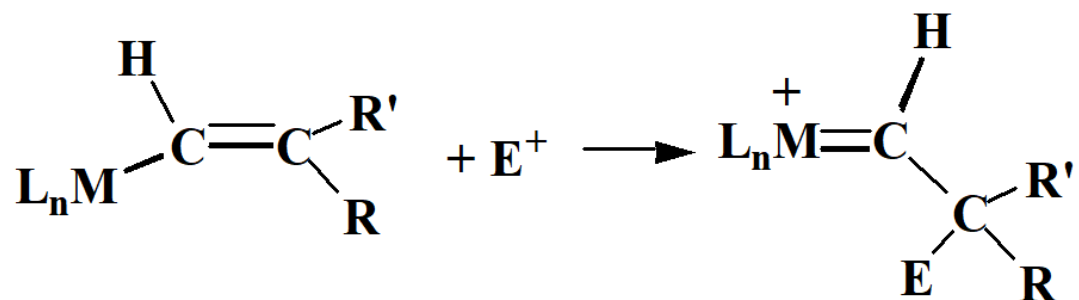
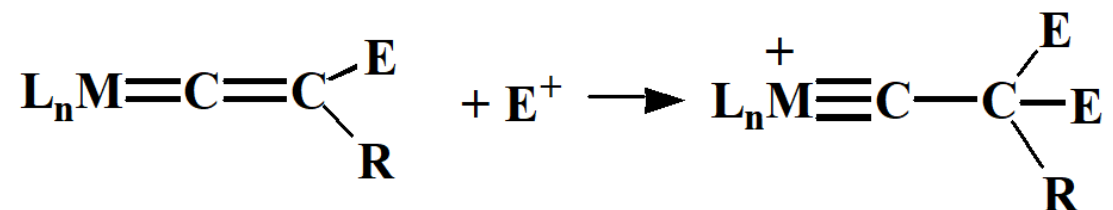
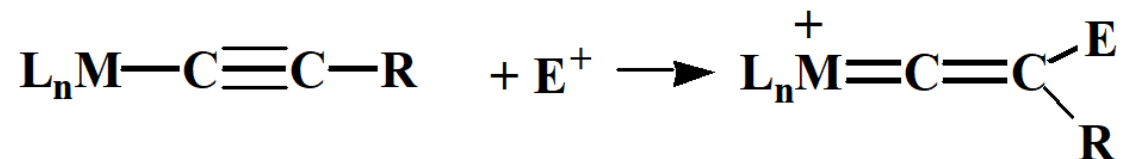


(2) Electrophilic addition to ligands.



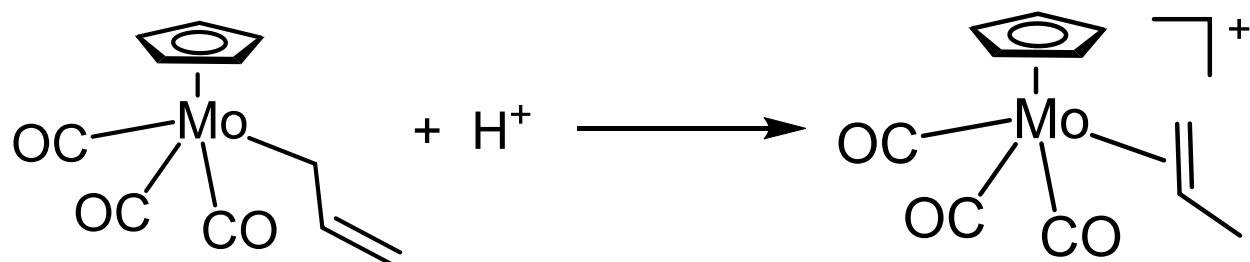
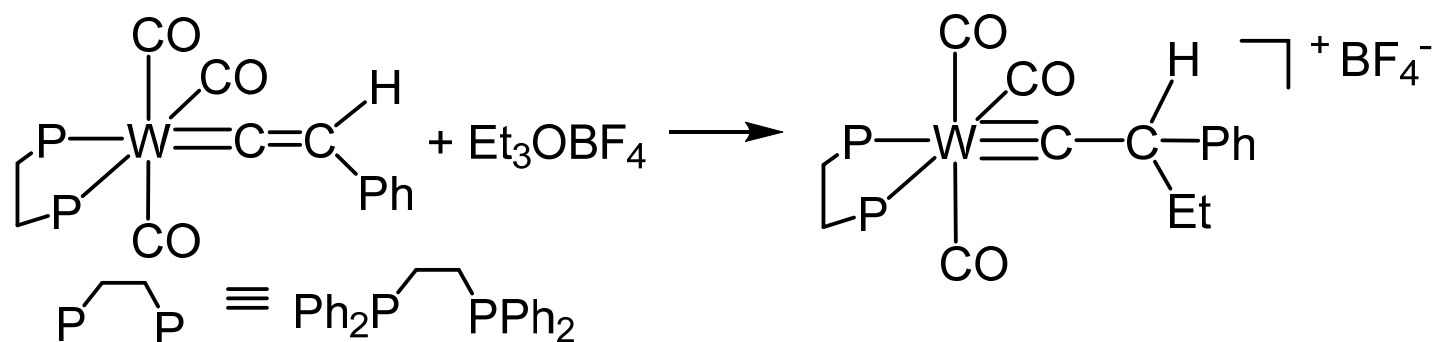
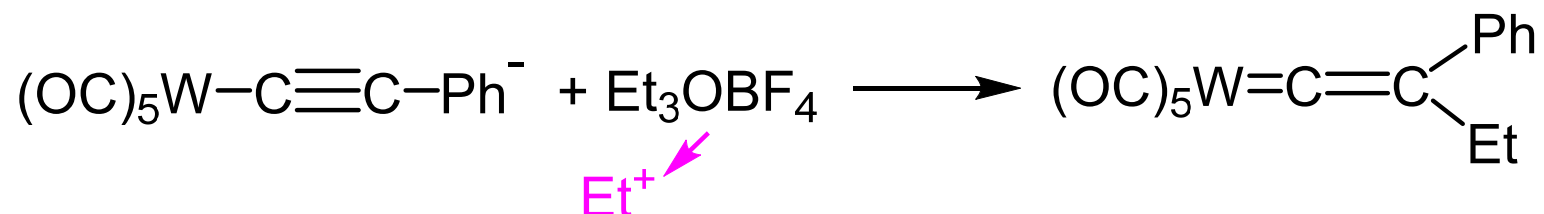
(Can be used for the synthesis of Fisher carbyne)

More examples



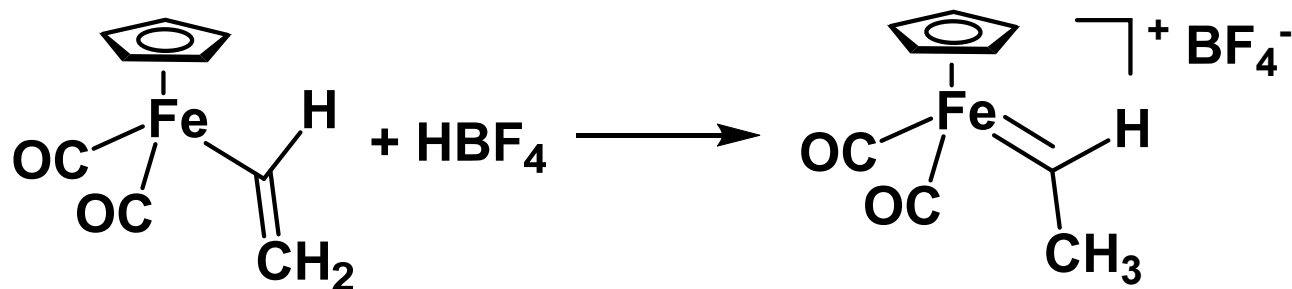
Practice Problems

Give the products of the following reactions.

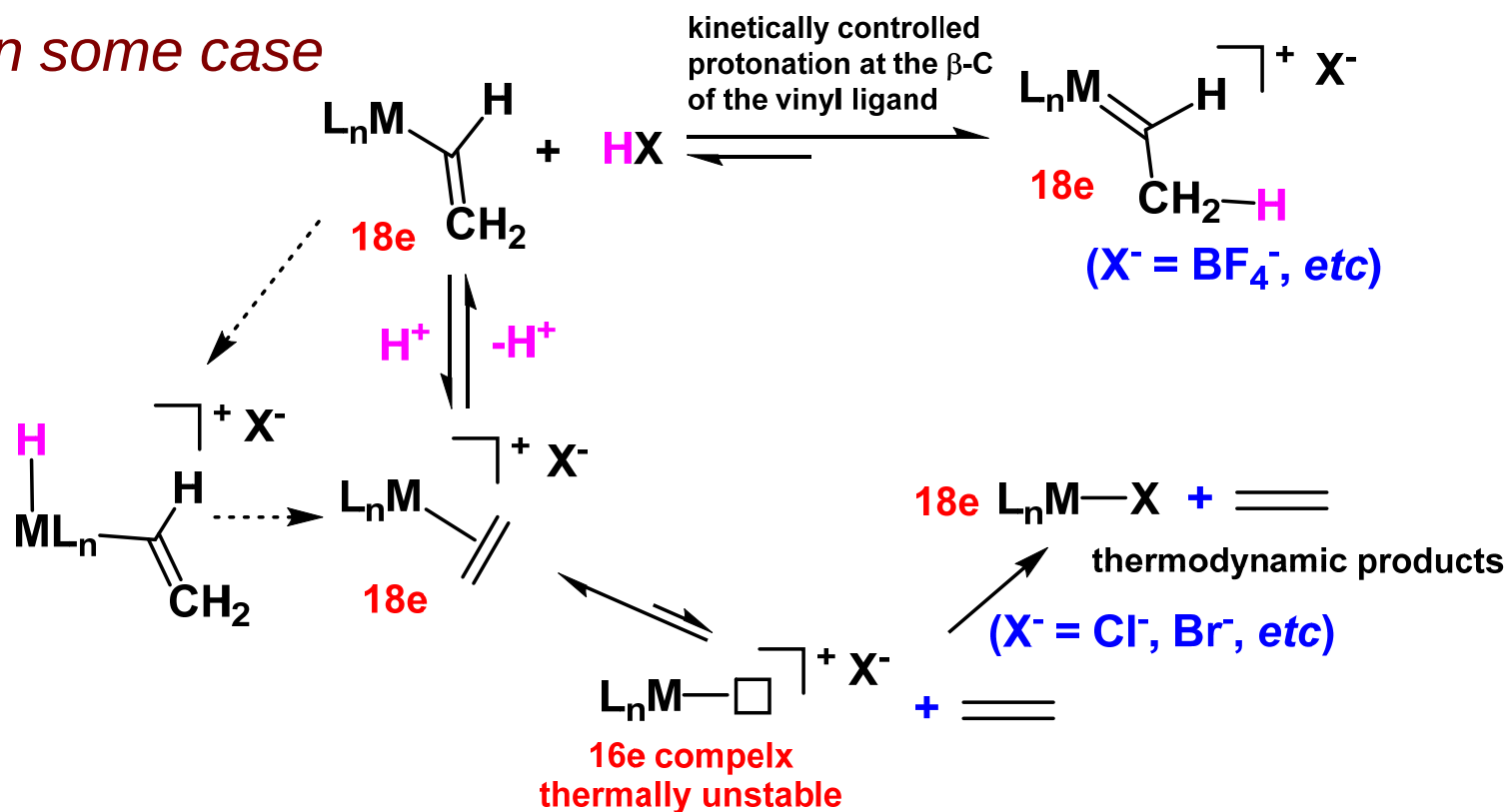


Practice Problems

Give the products of the following reactions.

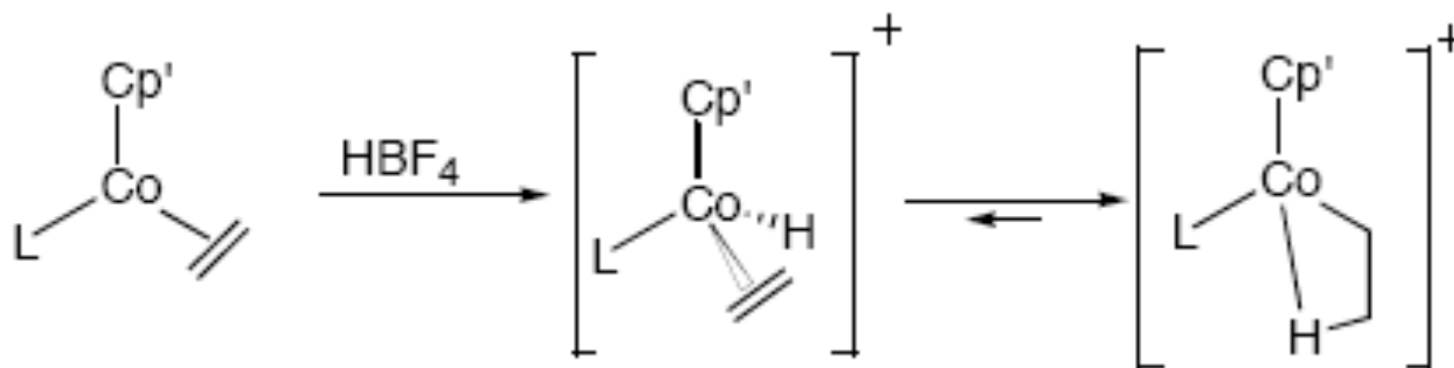


Note: in some case



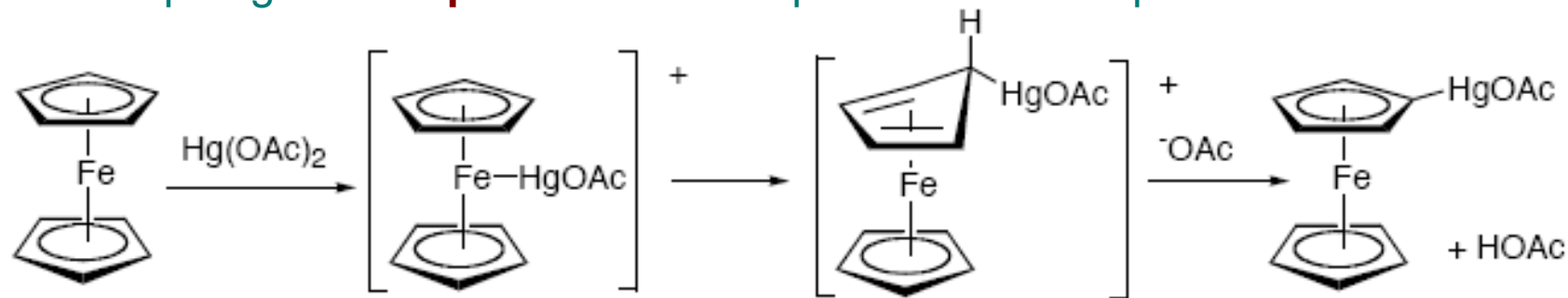
Comments on electrophilic addition reactions

Attack on olefin. The electrophile can either add first to the metal center, or directly to the ligand. The order of addition will depend on the relative basicity of the metal and the ligand.

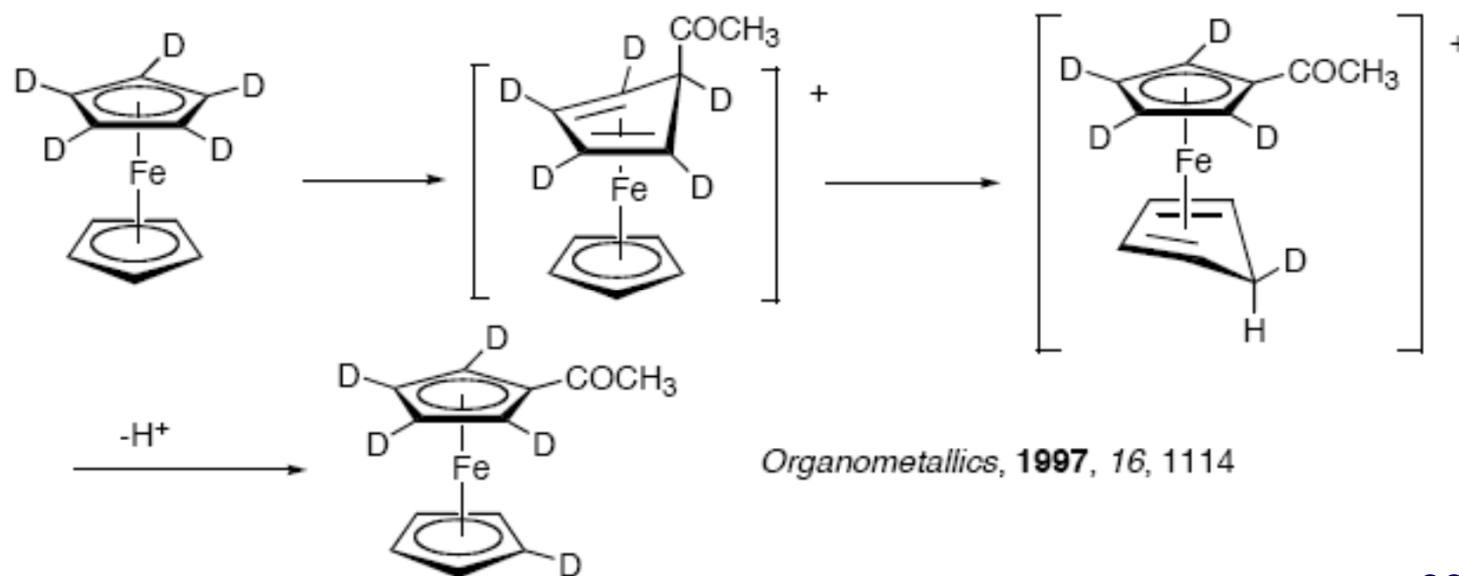


Electrophilic substitution of Cp ligand. Depending on the electrophile, addition to a Cp ring of ferrocene can either occur directly or by initial attack at iron.

Mercuric acetate adds **initially to the iron center** followed by **an endo** attack on the Cp ring. **Exo deprotonation** completes the electrophilic substitution.

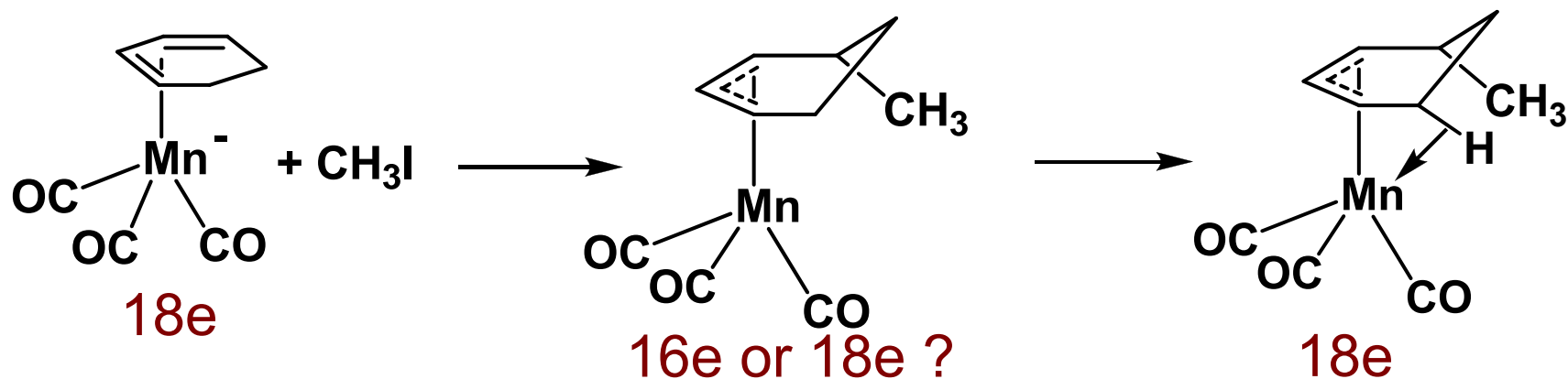
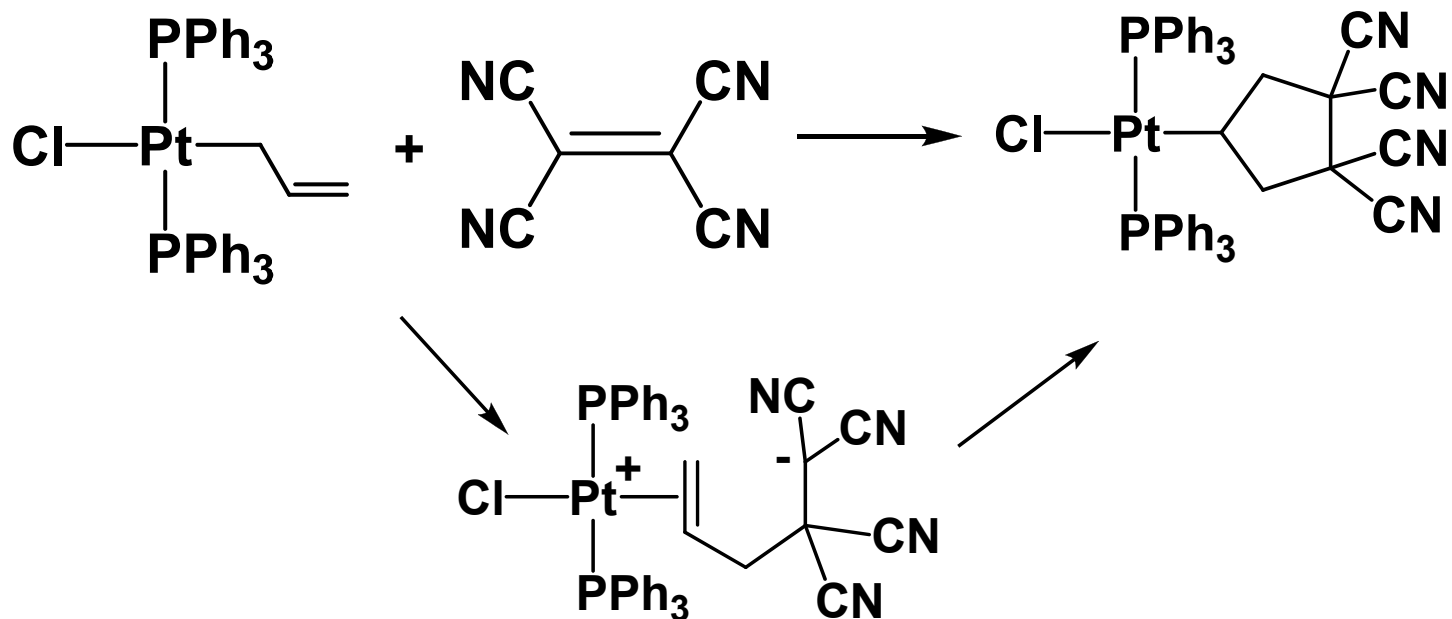


For a **hard electrophile**, such as CH_3CO^+ , **exo attack on the Cp ring** occurs directly

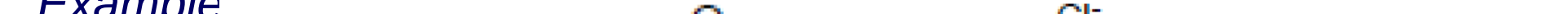


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The initially formed complex may undergo further transformation.
e.g.



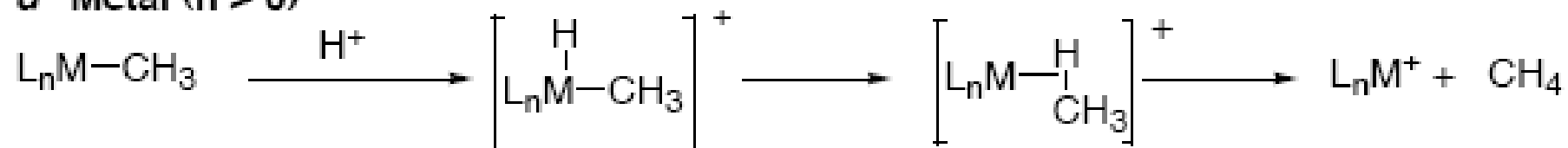
+



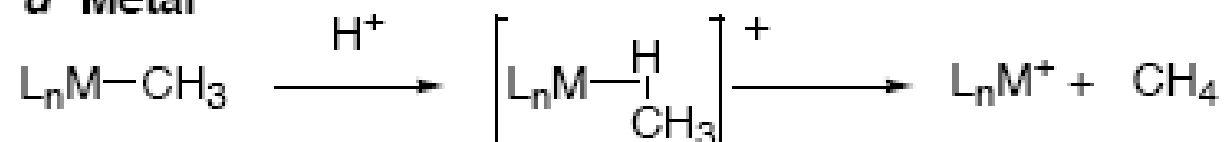
40

In general, for metals with d electrons, protonation at the metal followed by reductive elimination is thought to occur. Since d^0 metals cannot be protonated, attack must occur at the ligand.

d^n Metal ($n > 0$)



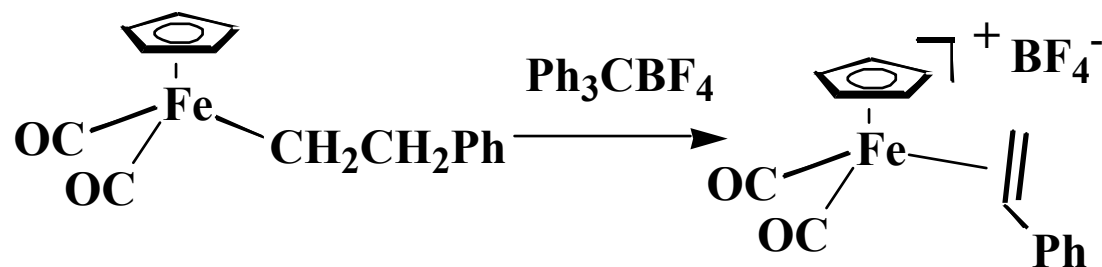
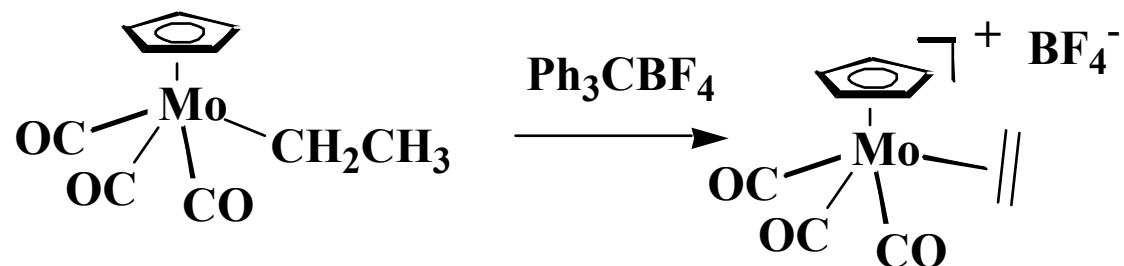
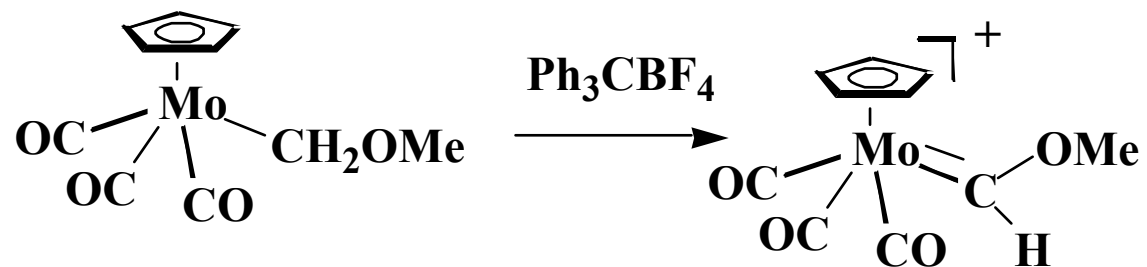
d^0 Metal

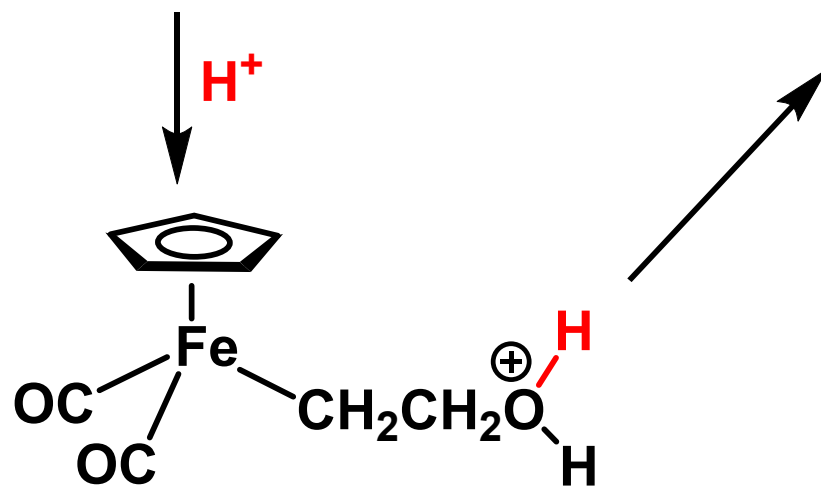
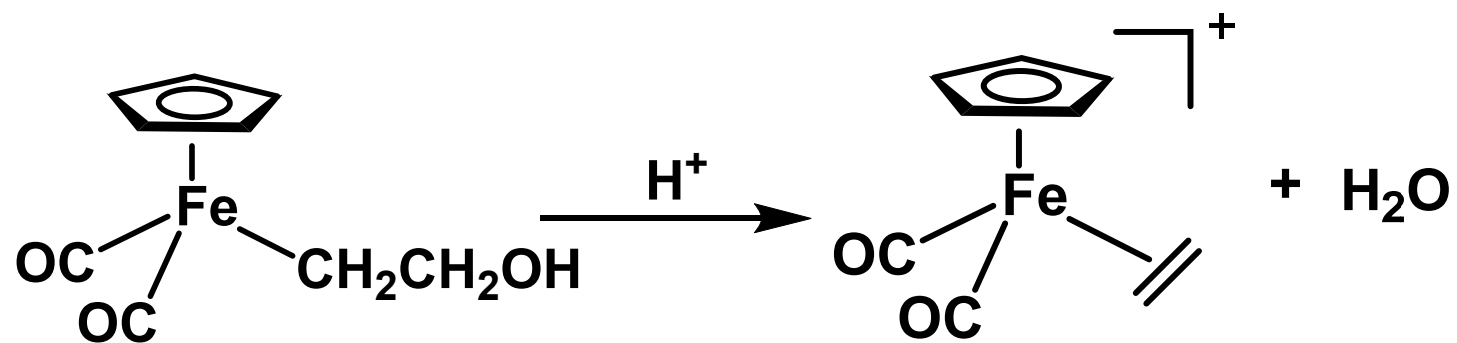
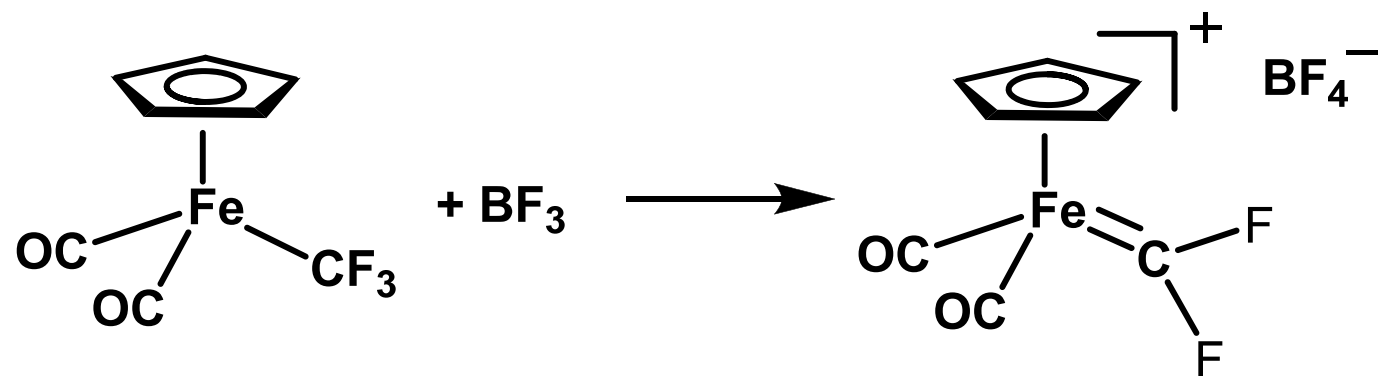


Early transition metals are very reactive towards even weak acids such as water, but later transition metals require more vigorous conditions.

(2) Electrophilic abstraction of other groups.

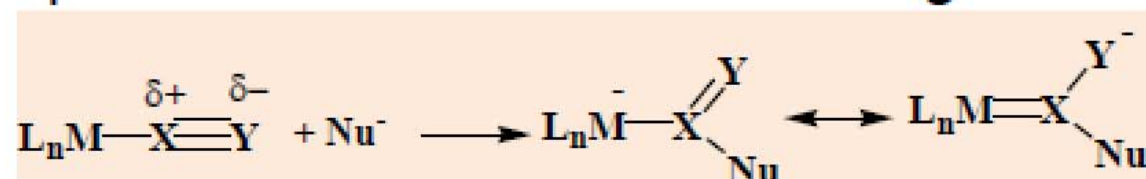
Some groups can be electrophilically abstracted. e.g.



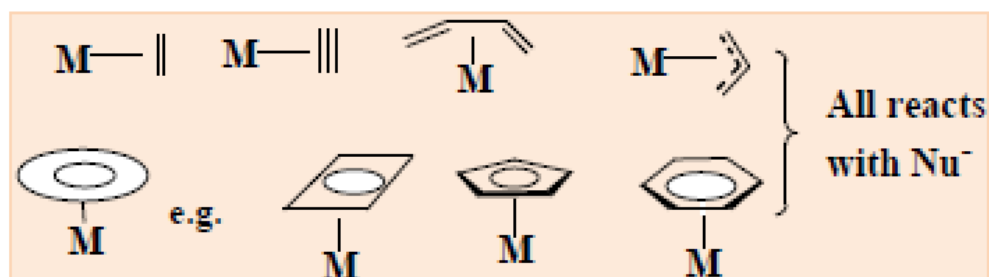


Summary

- Nucleophilic addition to CO and related ligands



- Nucleophilic addition to alkene and related ligands

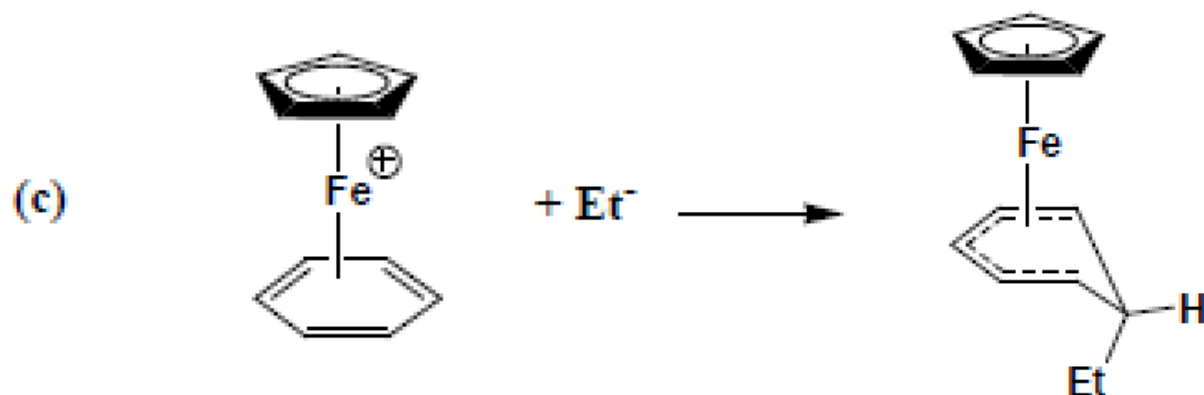
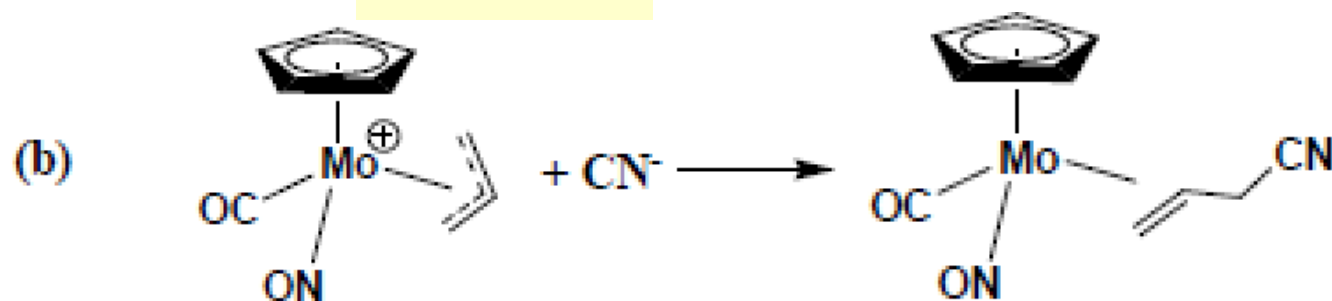
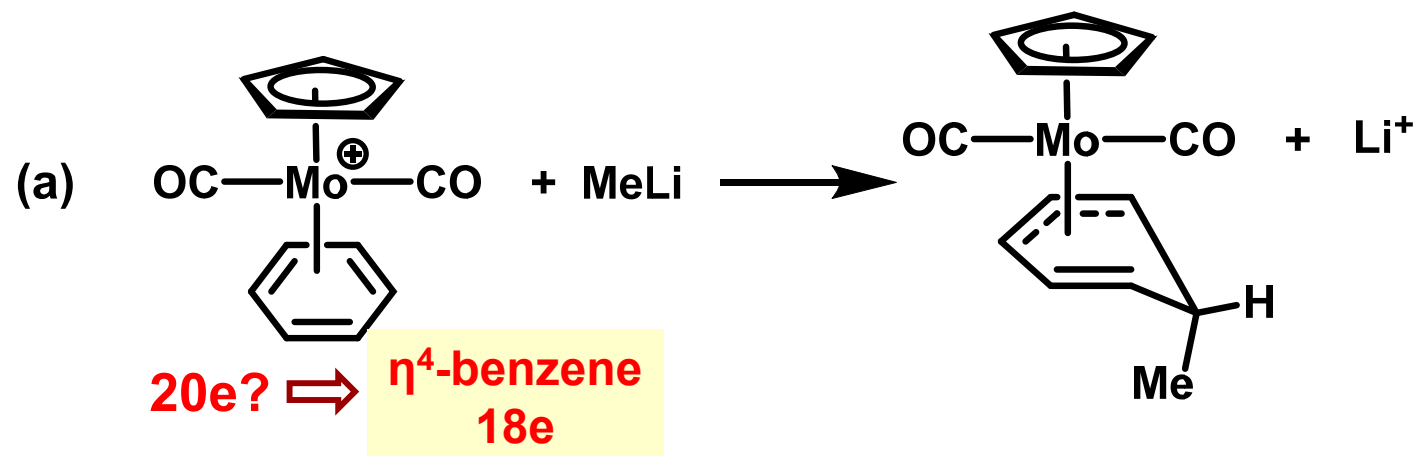


- Rules for
- Site selectivity
- Regio chemistry
- stereoselectivity

- Nucleophilic abstraction: deprotonation reactions
- Electrophilic addition/abstraction

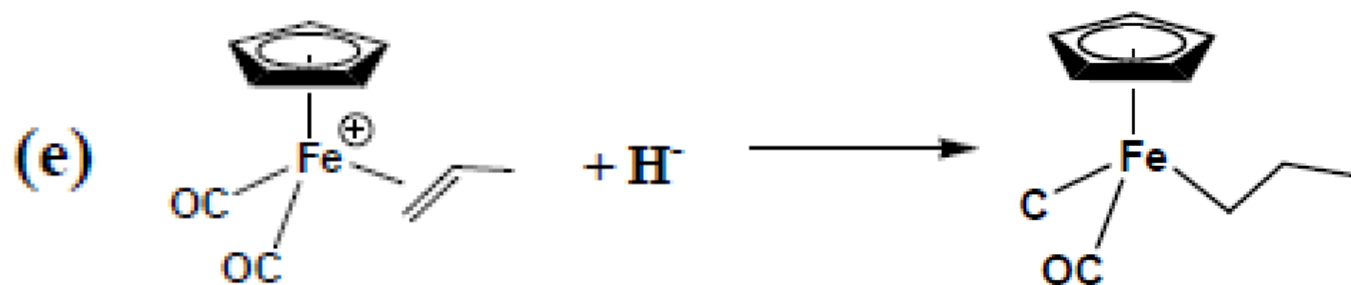
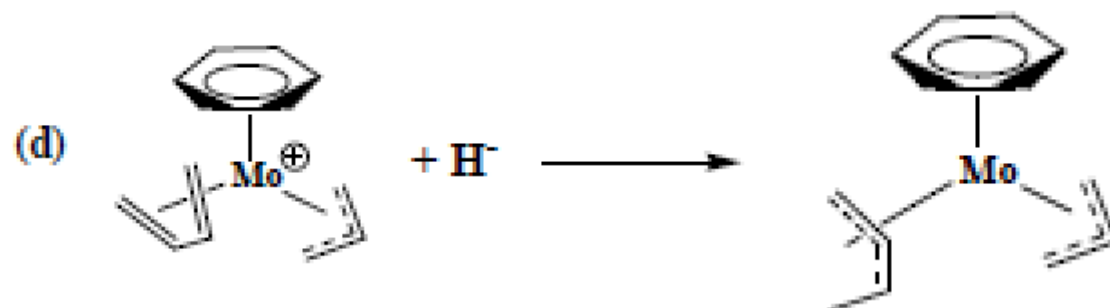
Review Questions

1. Give the products of the following reactions.



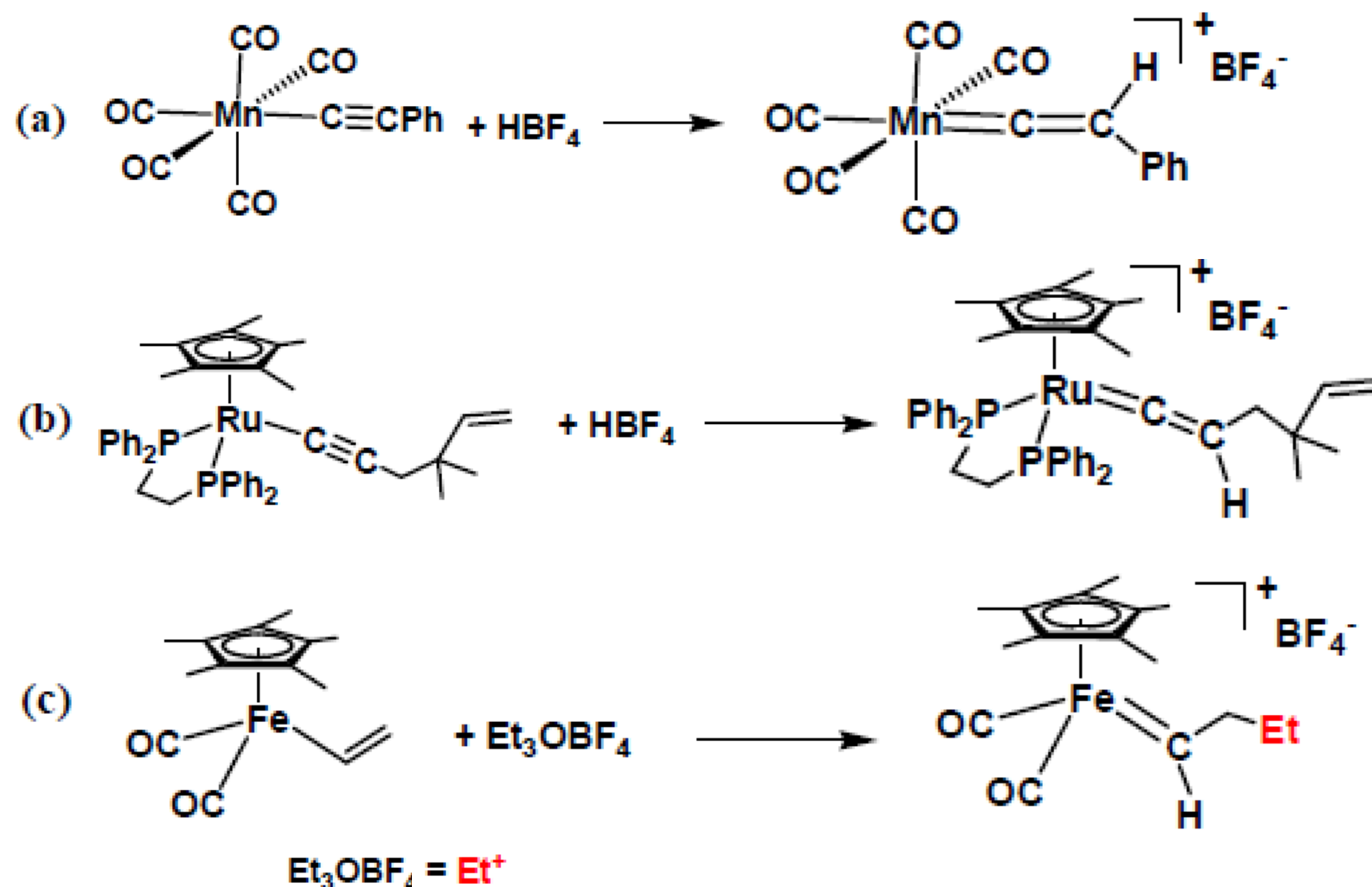
Review Questions

1. Give the products of the following reactions.



Review Questions

2. Give the products of the following reactions.



Suggested Reading: Facile Regio- and Stereoselective Hydrometalation of Alkynes with a Combination of Carboxylic Acids and Group 10 Transition Metal Complexes: Selective Hydrogenation of Alkynes with Formic Acid

Ruwei Shen,[†] § Tieqiao Chen,^{†,‡} § Yalei Zhao,[‡] Renhua Qiu,[‡] Yongbo Zhou,[‡] Shuangfeng Yin,^{*,‡} Xiangbo Wang,[†] Midori Goto,[†] and Li-Biao Han^{*,†,‡}

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